Microcontroller Based Systems & Robotics Hub

Promoting robotic design and entrepreneurship experiences among students and teachers

Lesson 4:
Basic Electronics

CONTENTS



- Breadboarding circuits
- What is electricity
- Voltage, current, resistance
- Ohm’s law, circuit basics
- Electronic components
- Analog and digital signals
- Digital electronics (logic gates)

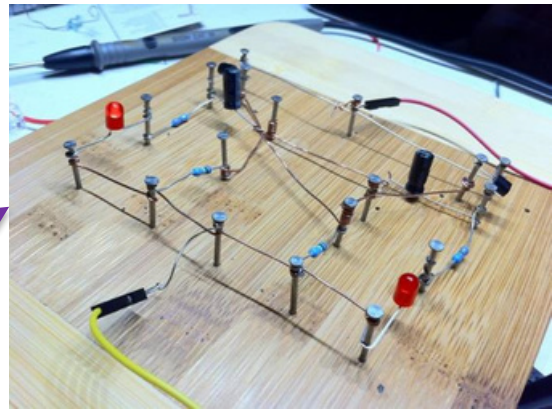
THE BREADBOARD

- A breadboard is a thin-plastic board used to hold and connect electronic components and/or wires temporarily
- It is used for circuit prototyping, i.e., building an actual circuit to verify a theoretical circuitdesign

Years ago:

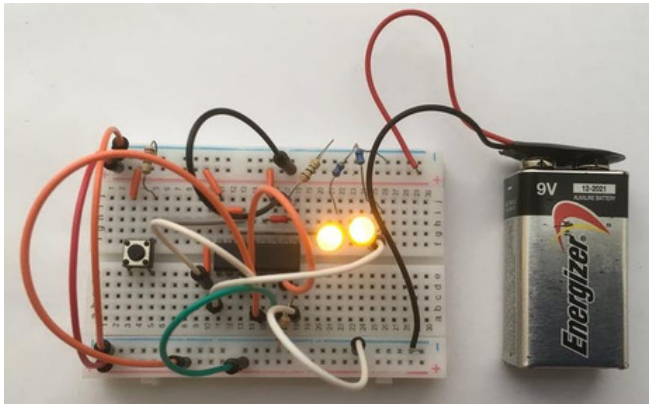


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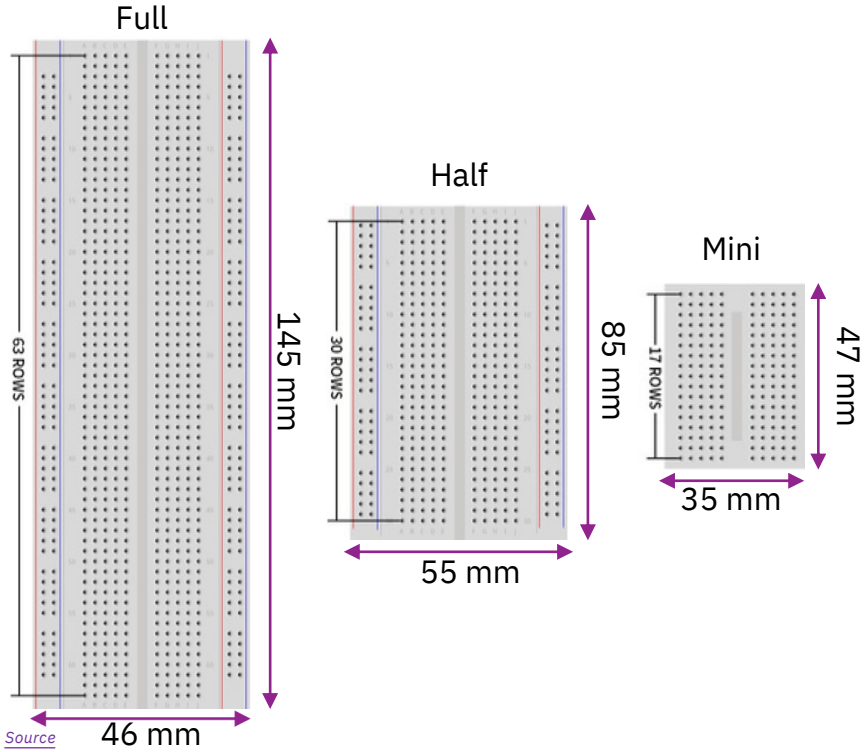
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Now:



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THE BREADBOARD



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Common sizes of breadboards

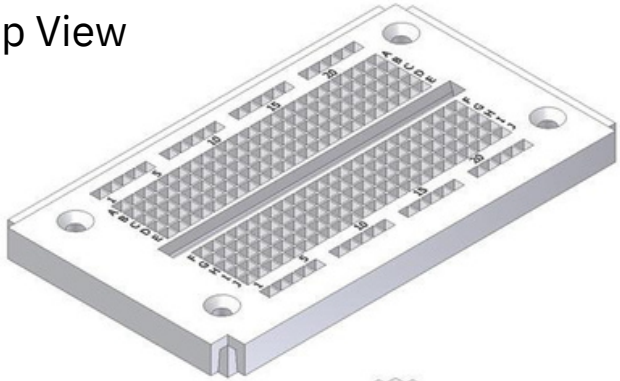


Video

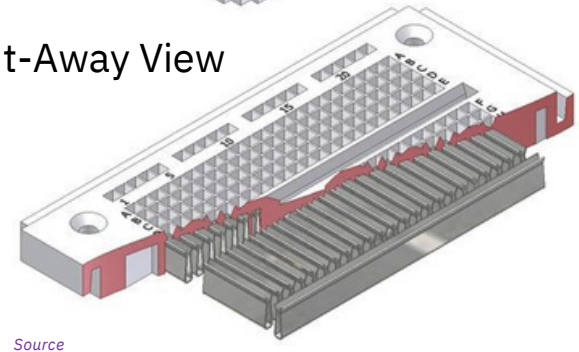
HOW A BREADBOARD WORKS

- The surface of the breadboard has holes arranged in a grid pattern
- Electric component leads and the wires used to connect them are inserted into these holes
- A series of internal metal strips serve as jumper wires, they connect specific rows of holes

Top View



Cut-Away View

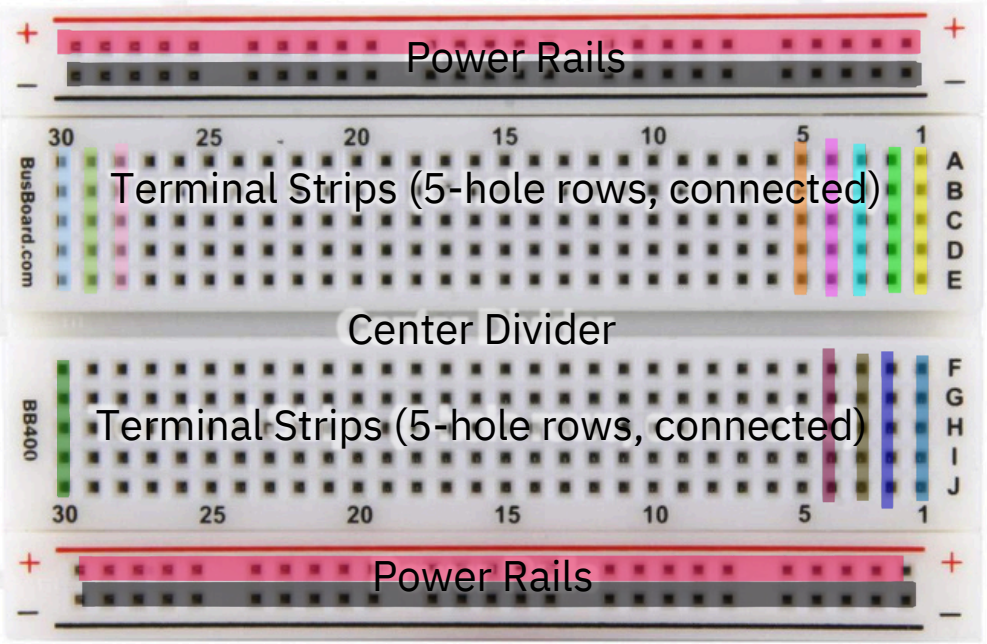


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BREADBOARD CONNECTIONS

1-30 are connected (even along the breaks, for example, between 5 and 6)

A-E and F-J are connected internally (but not across the middle divider, i.e., not between E&F)

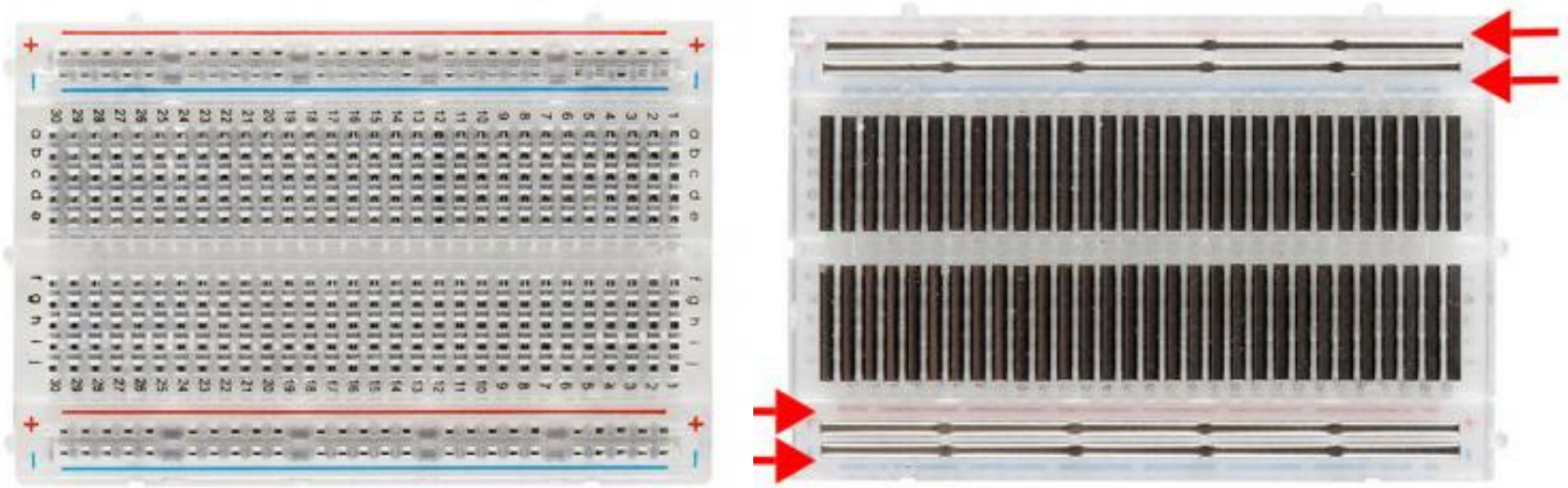


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NOTE : Same colored rows are connected internally

BREADBOARD CONNECTIONS

Inside a breadboard

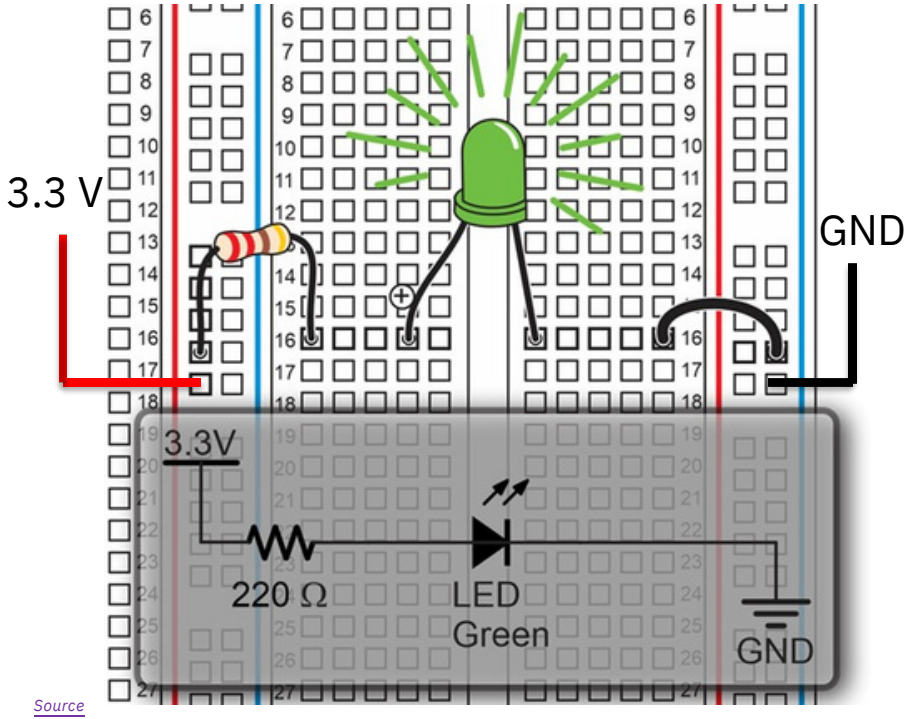


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BREADBOARD: GUIDELINES AND TIPS

To make troubleshooting easier:

- Use the breadboard to connect majority of the components
- Use as few jumper wires as possible
- Keep jumper wires short to help avoid a jumble of wires
- Make connections to resemble the circuit schematic as close as possible

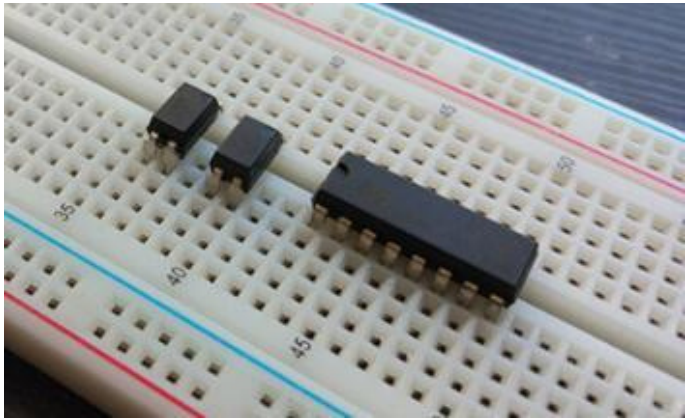


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Breadboard circuit ~ schematic circuit

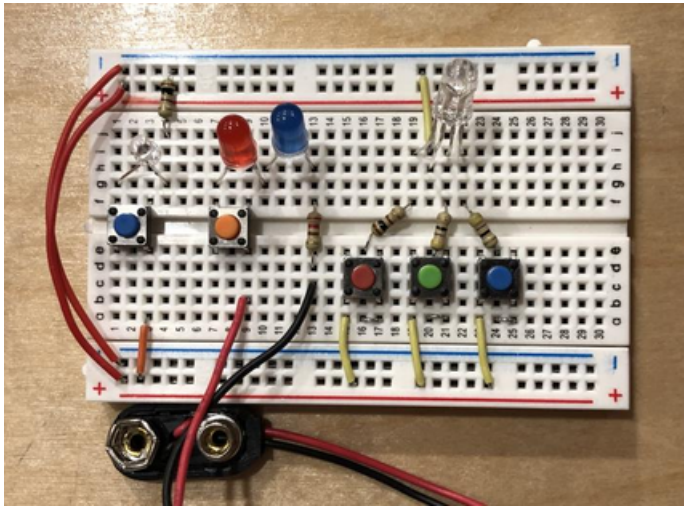
BREADBOARD: GUIDELINES AND TIPS

- IC chipsmust be the middleof the breadboard, as shown below.



Source

- Cut component leads to manageable lengths, to avoid short circuit and maintain a clean circuit.

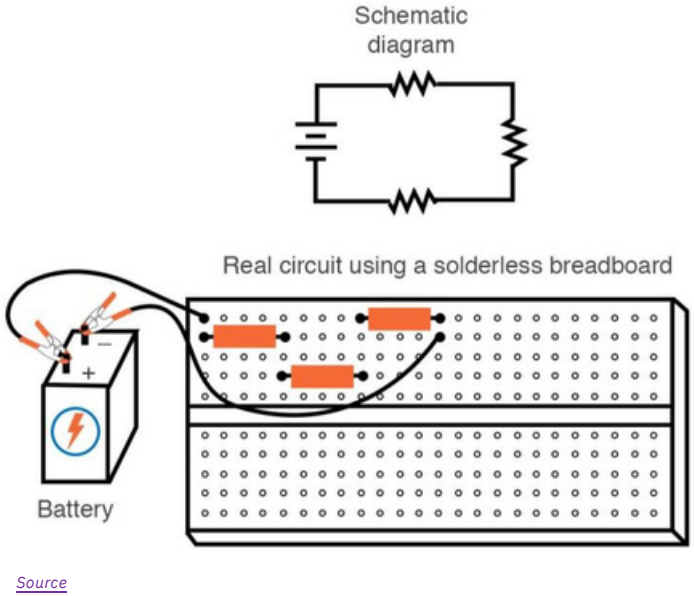


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BREADBOARD: GUIDELINES AND TIPS

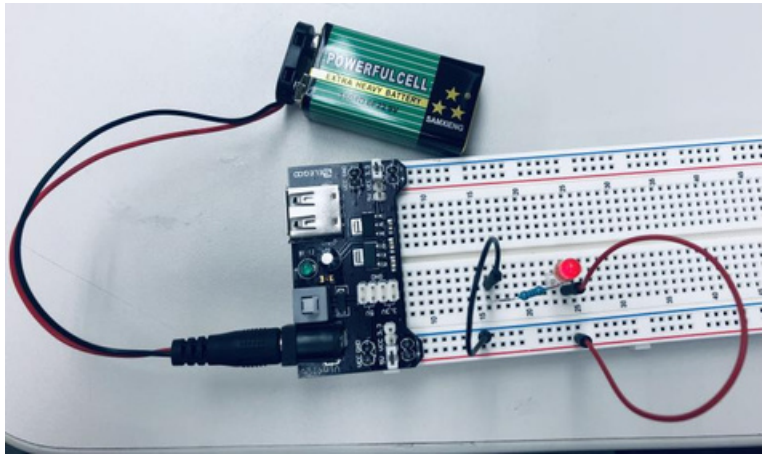
- Work from a circuit schematic and check off the connections on the breadboard one by one

- Have someone check your circuit for errors



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POWER SUPPLY MODULE



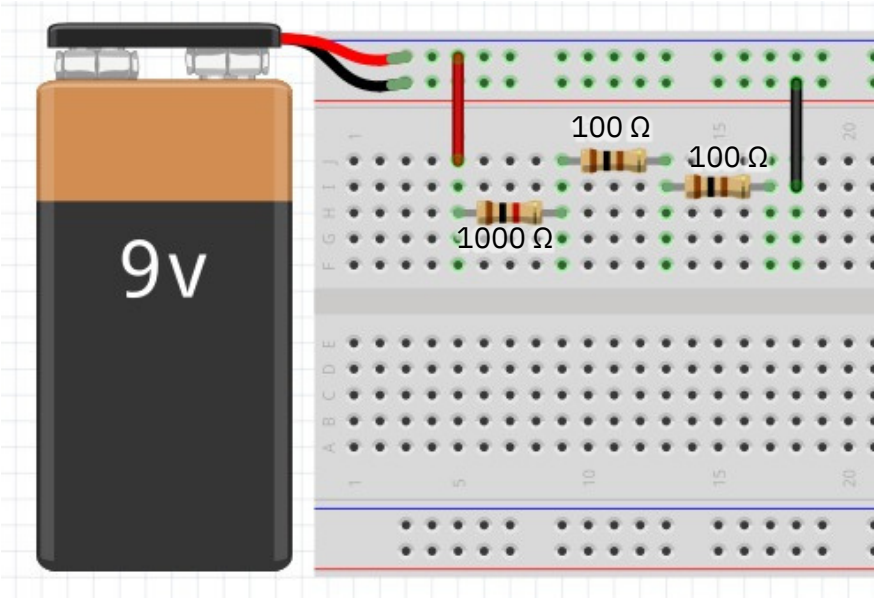
Input Voltage : DC 6.5 -12V

Output Voltage : DC 3.3V -5V

Instead of directly powering the breadboard using a battery and connector (without barrel plug),

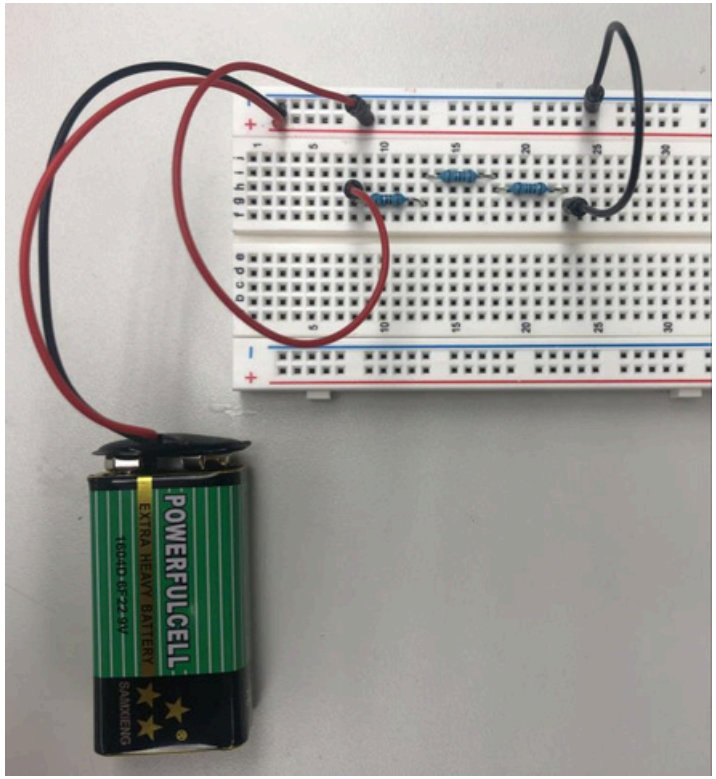
A power supply module can be used to supply a constant voltage of 5V to the breadboard using a battery connector (with barrel plug)

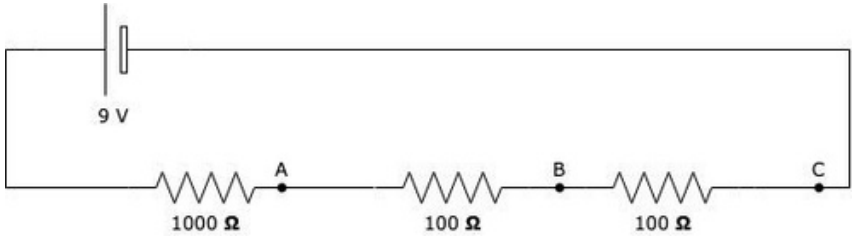
ACTIVITY -I



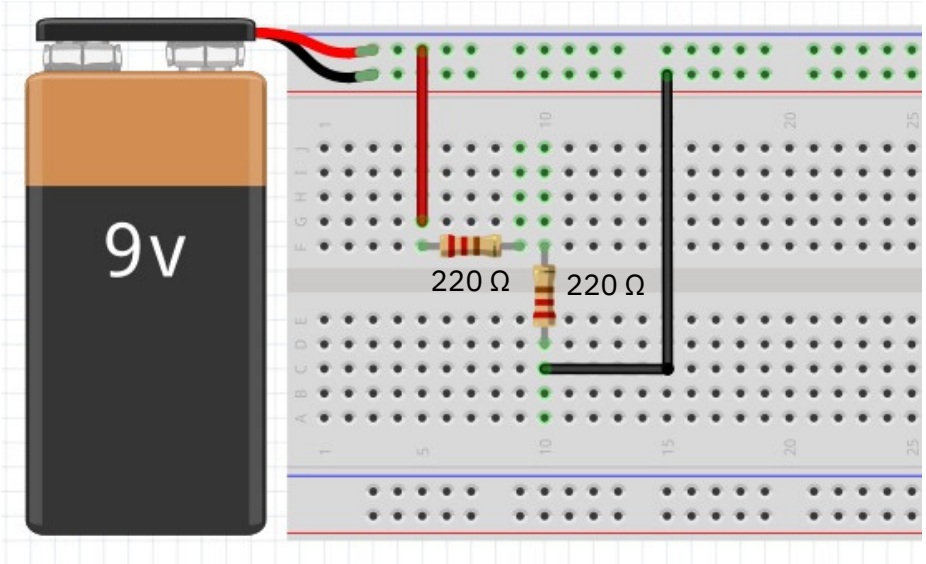
Construct the circuit above using guidelines for proper breadboarding

ACTIVITY -I (SOLUTION)



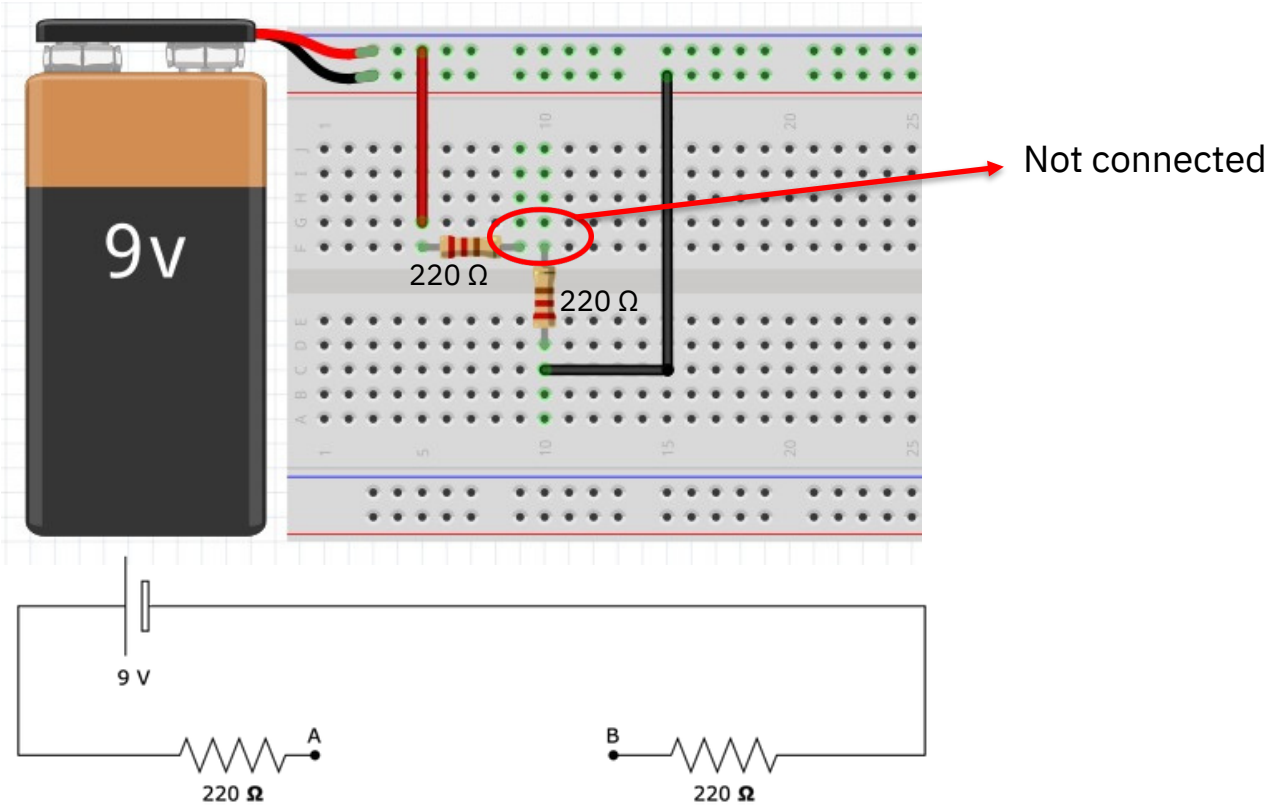


ACTIVITY -II



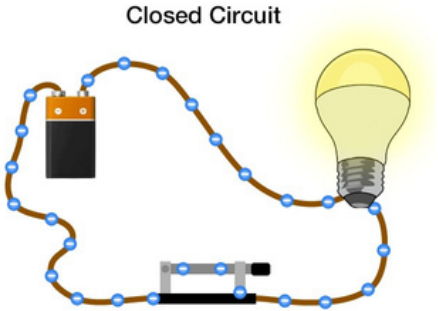
Determine the error in this circuit

ACTIVITY –II (SOLUTION)

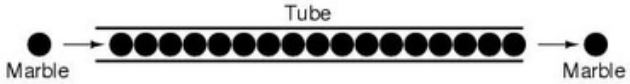


WHAT IS ELECTRICITY

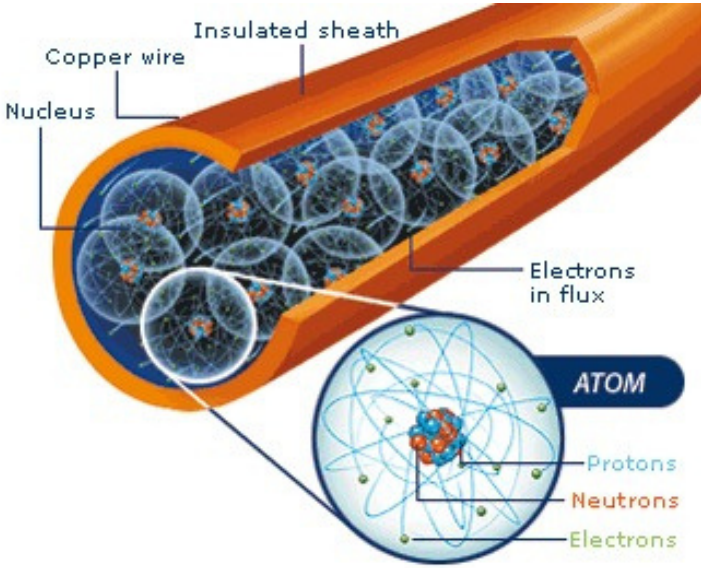
- Everything is made of atoms
- Atom consists of electrons, protons, and neutrons
- Flow of electrons is called electricity



Electric Circuit



Speed of electricity (learn more)

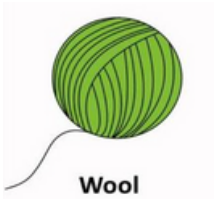


Electrons in a copper wire (Not to scale)

CONDUCTORS AND INSULATORS

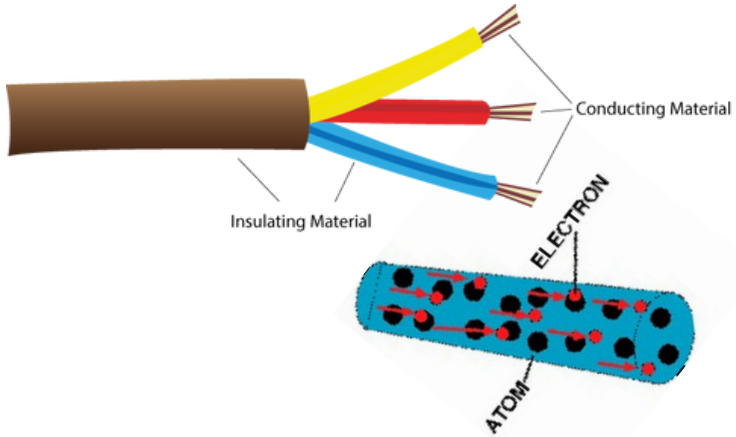
Insulators: Some materials have strong attraction forces and refuse to lose electrons, these are called insulators (air, glass, rubber, most plastics)

Conductors: Some materials have weak attractions and allow electrons to be lost, these are called conductors (copper, silver, gold, aluminum)

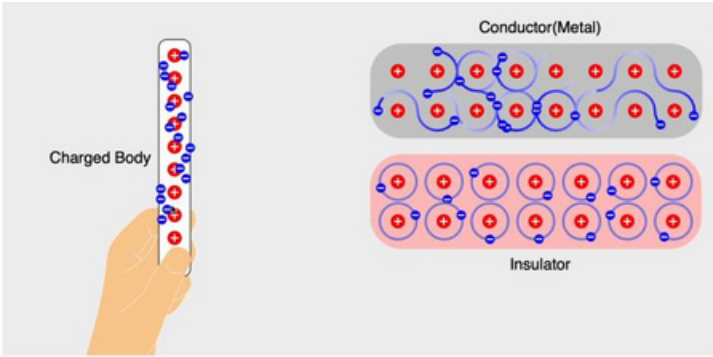




Insulators

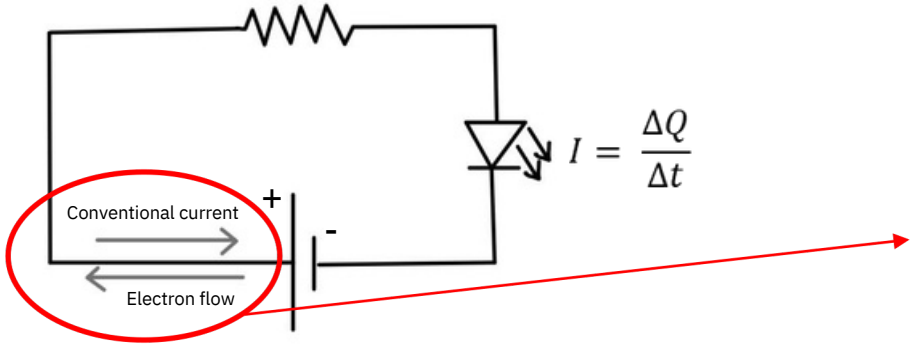


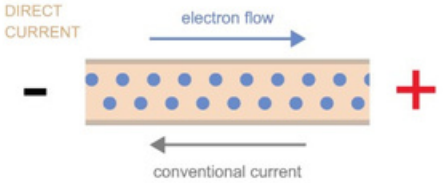
Conductors



Concept of free electrons → (Try Yourself)

ELECTRIC CURRENT



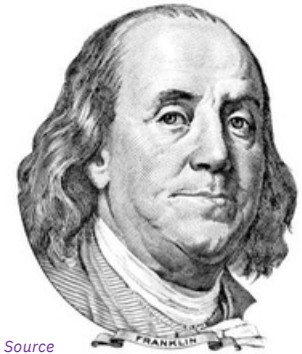


Electron flow v/s coventional current

- Electric current is the flow of charge
- ΔQ is the amount of charge passing through a conductor at any location during some interval of time ΔT
- SI units of current: Amperes (A) 1A = 1 Coulomb/second
- 1.602 x 1019electrons/sec = 1A current
- Charge carriers are the electrons which flow out of the (-) terminal of the battery and into the (+) terminal

CURRENT DIRECTION: CONVENTIONAL

In the 1700s, Benjamin Franklin considered electricity analogous to an invisible fluid



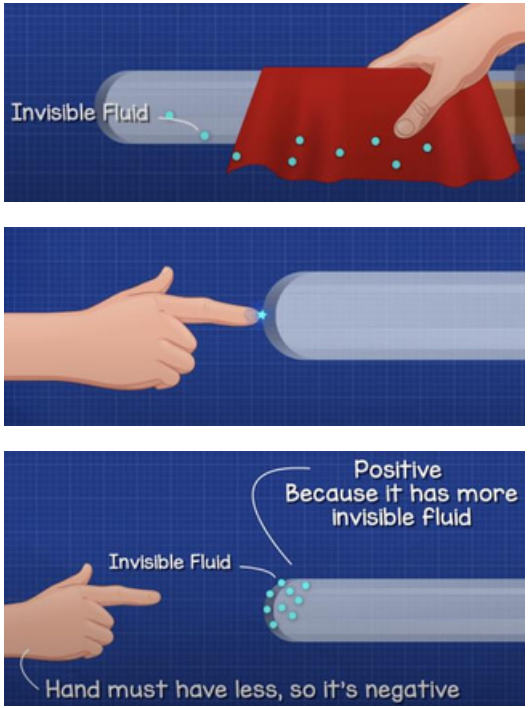
Source

Experiment:

- Rubs glass rod with silk cloth
- Human hand touches the glass rod, observes a spark (now known as static electricity)

Conclusion:

- Assumed that currentflows from positive(glass rod) to negative side (hand)
- Similar to how water flows from high level to low level



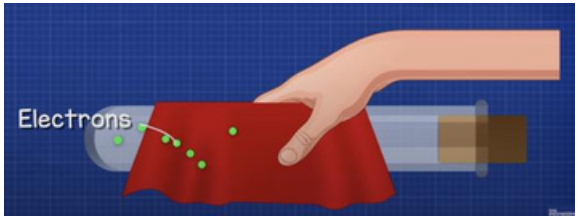
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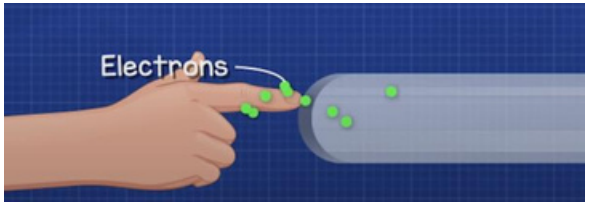
CURRENT DIRECTION: ELECTRON FLOW

In reality:

- Silk cloth removes the electrons from the glass rod
- Human hand provides electrons to the glass rod
-

  The electrons flow from negative to positive side, i.e., the human hand is negatively charge and the glass rod is positively charged
  Opposite of what was assumed by Benjamin
- Franklin





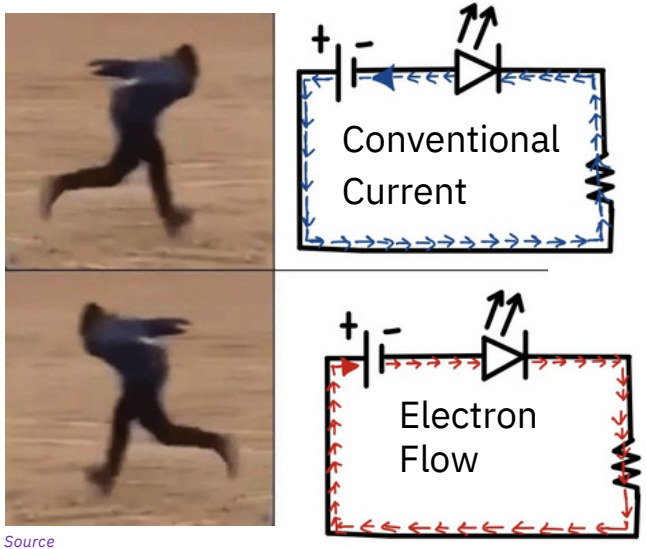
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But ‘Electron ’ was only proven to exist by Joseph Thomson in 1897

CURRENT DIRECTION: CONCLUSION



Source

Ideally:



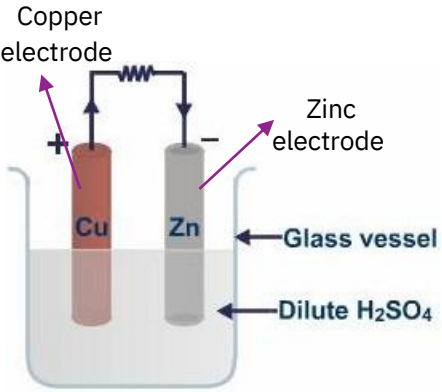
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It should be known that electrons being the charge carriers, the actual current direction is the that of electron flow

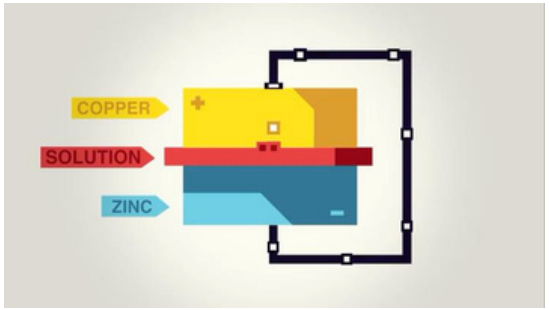
However, most textbooks and equations in the electrical domain assume conventional current direction

BATTERIES

- A battery produces electricity by transforming chemical energy into electrical energy, using various metal electrodes immersed in an electrolytic solution
- Batteries are the source of potential difference $\Delta V = V_a - V_b$ and measured in SI units of volts (V)



Source



How batteries work



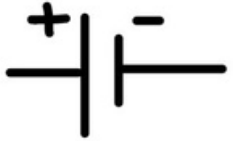
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Source

Alessandro Volta - inventor of electric cell

Circuit symbol of battery:

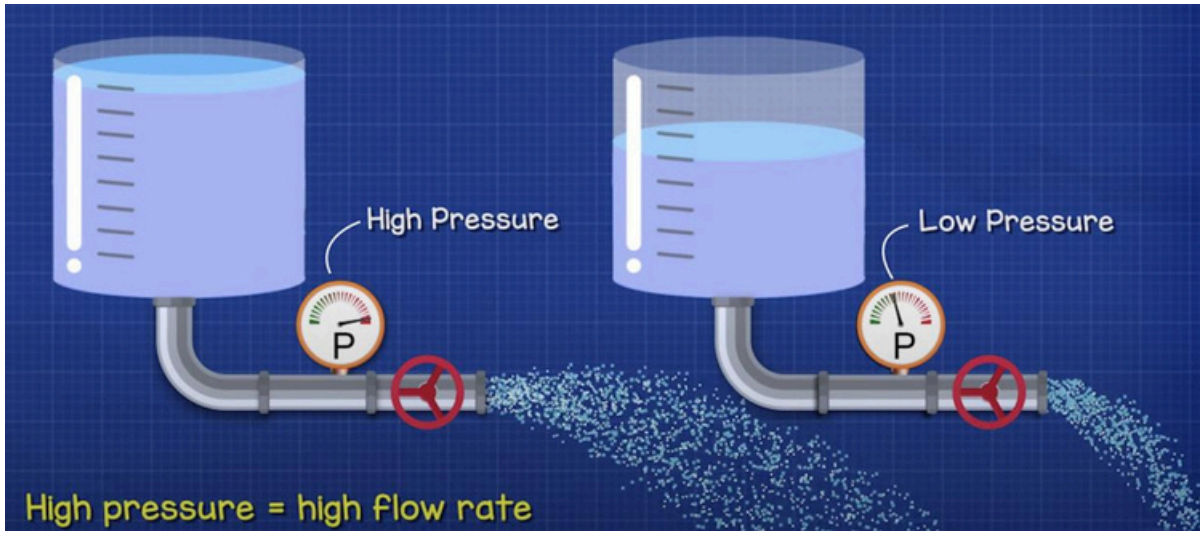


Is there a simpler way to understand current and voltage?

VOLTAGE AND CURRENT

Voltage → Pressure

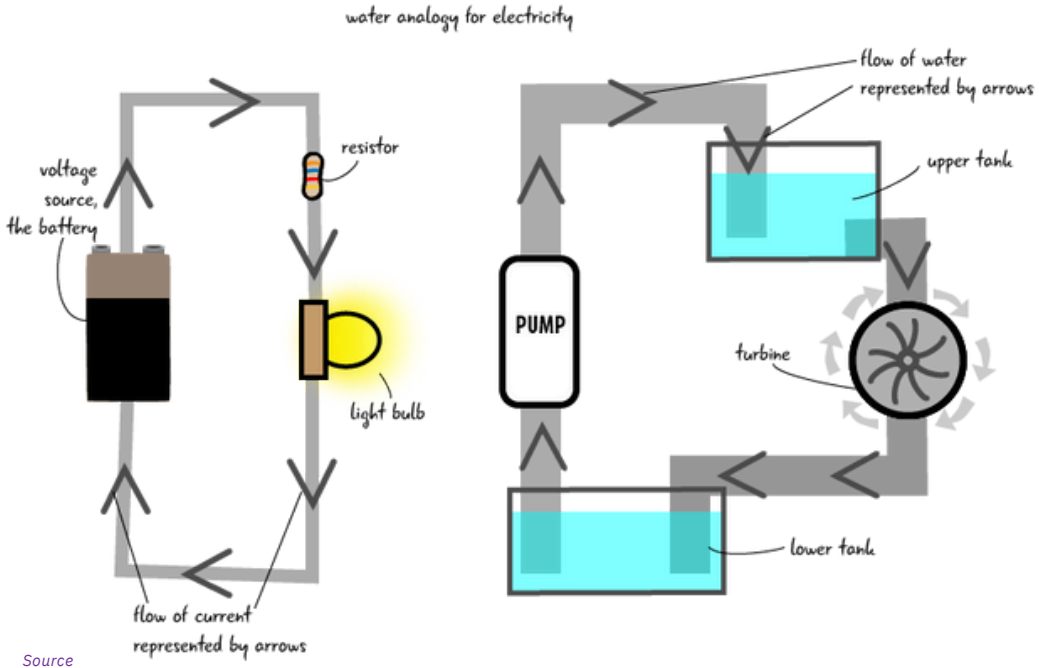
Current → flow of water



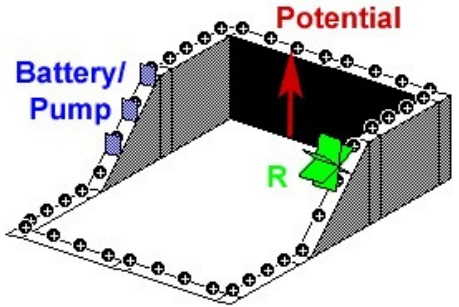
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What kind of connections do we have for flow of electricity?

VOLTAGE AND CURRENT



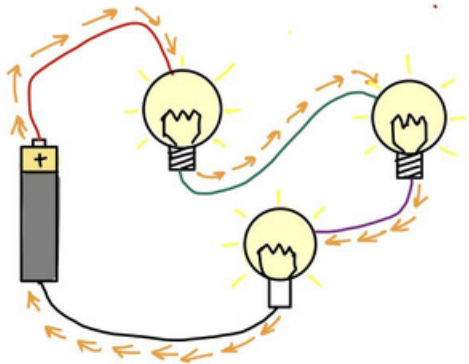
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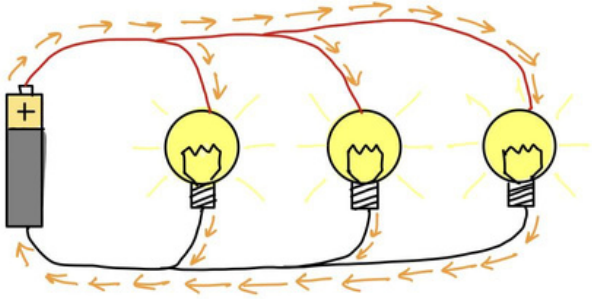


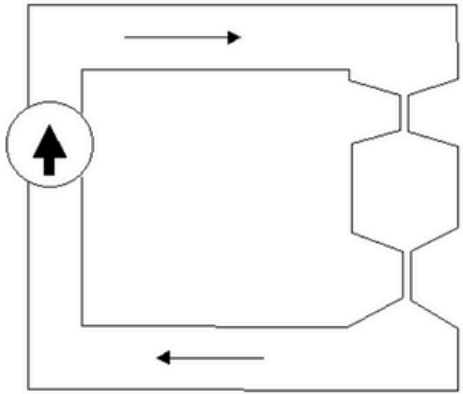
Water analogy

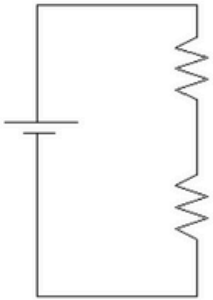
What kind of connections do we have for flow of electricity?

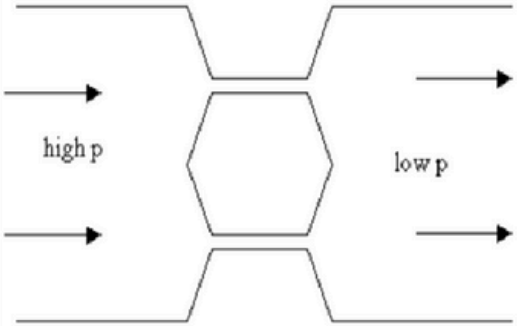
SERIES VS. PARALLEL CIRCUITS

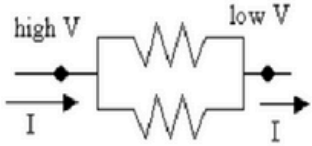








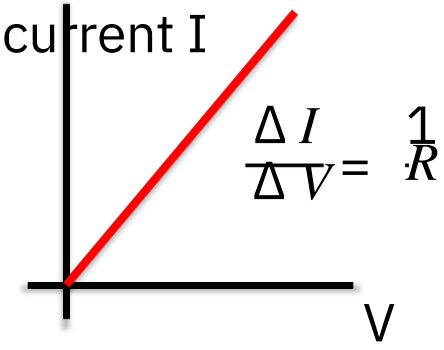




What about components other than batteries and bulbs?

RESISTANCE

In 1850, Georg Simon Ohm determines that flow of electric current I through a conductor experiences a certain amount of resistance

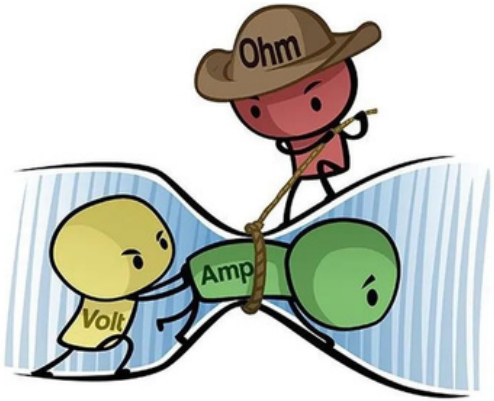


The resistance expressed in ohms, named after George ohm, is a measure of how much. A resistor resists the flow of electricity
This impediment to current flow is due to electron collisions resulting in behavior called resistance

$$I \propto V \rightarrow I = G \times V \rightarrow I = \frac{V}{R} \rightarrow R = \frac{V}{i} \rightarrow V = I \times R$$

V=IxRis called Ohm's law
R is the resistance and G is the conductance in the circuit

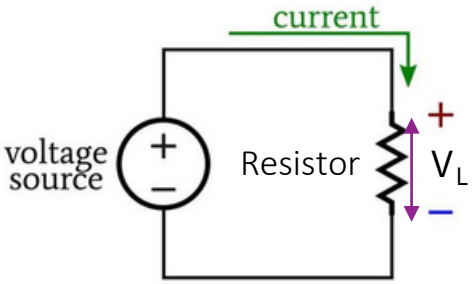
SI unit: Ohms, 1 Ω= 1 volt /ampere



Source

RESISTORS

- All electrical components which represent a loadin a circuit can be modeled as an impedance(a general form of resistance)

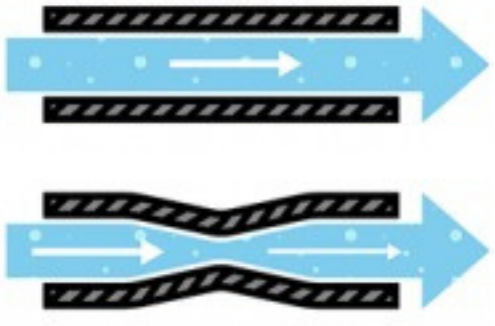


Source

- A *resistor* is a device that offers a specified amount of resistance and is used to control current in an electronic circuit



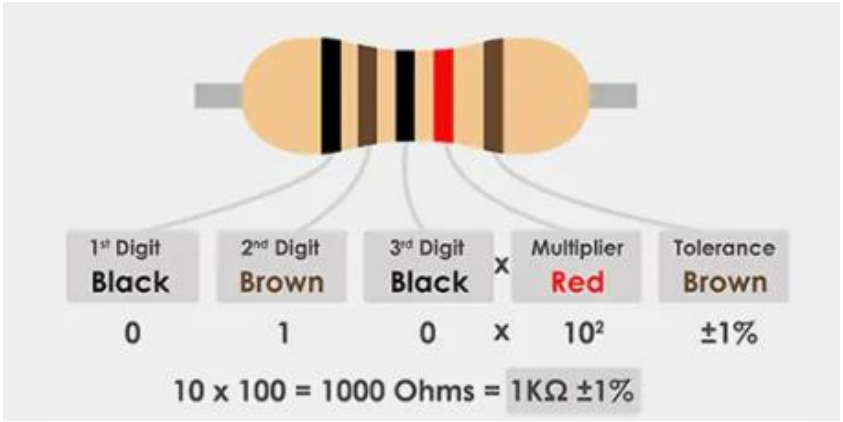
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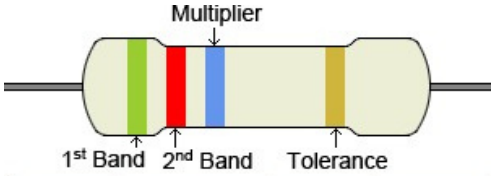
Source

RESISTOR COLOR CODING

- Resistors are color coded by various bands to indicate the resistance value.
  The number of bands are usually 4 or 5.
-
- The resistance value is calculated as shown



Source



Color	1st, 2nd Band Significant Figures	Multiplier	Tolerance
Black	0	× 1	
Brown	1	× 10	±1% (F)
Red	2	× 100	±2% (G)
Orange	3	× 1K	±0.05% (W)
Yellow	4	× 10K	±0.02% (P)
Green	5	× 100K	±0.5% (D)
Blue	6	× 1M	±0.25% (C)
Violet	7	× 10M	±0.1% (B)
Grey	8	× 100M	±0.01% (L)
White	9	× 1G	
Gold		× 0.1	±5% (J)
Silver		× 0.01	±10% (K)

Source

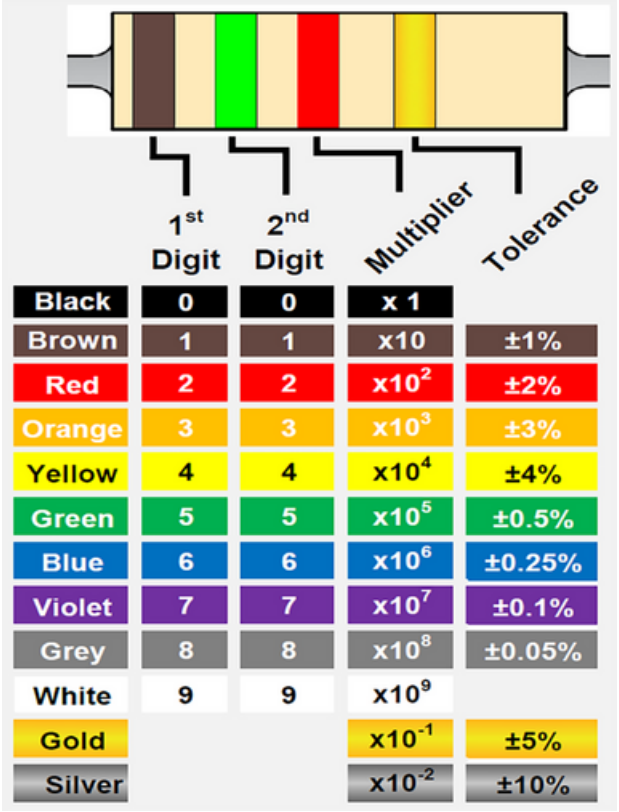
MULTIMETER



Video

ACTIVITY III

- Calculate resistor value based on color coding
- Check using multimeter



	1st Digit	2nd Digit	Multiplier	Tolerance
Black	0	0	x 1	
Brown	1	1	x10	±1%
Red	2	2	$x10^2$	±2%
Orange	3	3	$x10^3$	±3%
Yellow	4	4	$x10^4$	±4%
Green	5	5	$x10^5$	±0.5%
Blue	6	6	$x10^6$	±0.25%
Violet	7	7	$x10^7$	±0.1%
Grey	8	8	$x10^8$	±0.05%
White	9	9	$x10^9$	
Gold			$x10^{-1}$	±5%
Silver			$x10^{-2}$	±10%

Source

ACTIVITY III













- Calculate resistor values based on color coding
- Check values with multimeter readings

ACTIVITY III (SOLUTION)



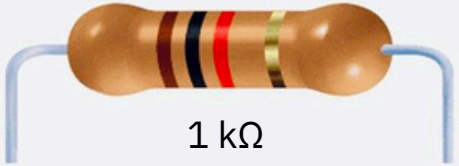
100 Ω



220 Ω



330 Ω



1 kΩ



100 Ω



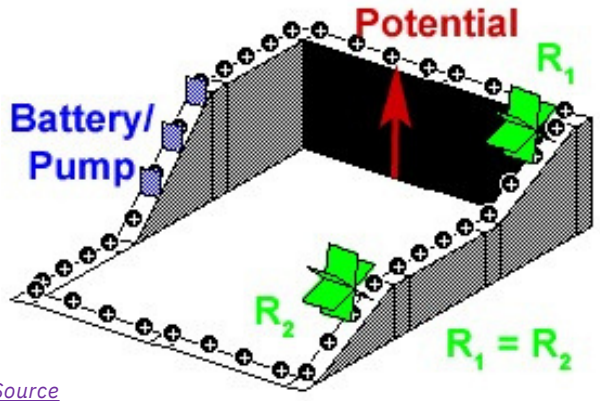
5.1 kΩ

RESISTORS IN SERIES

- When connected in series, the total resistance (RT) is equal to:

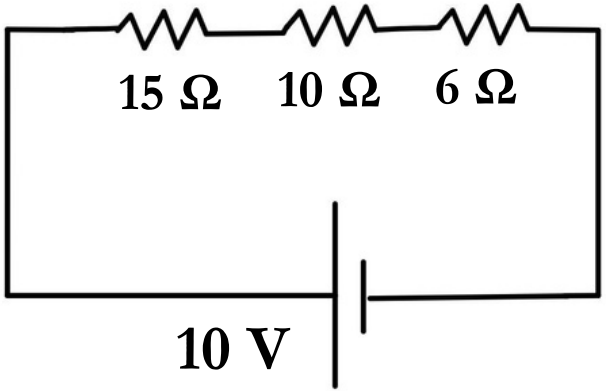
$$RT = R1 + R2 + R3 + \ldots$$

- The total resistance is always larger than any individual resistance.



Source

RESISTORS IN SERIES



Example :

$R_T = 15+10+6 = 31\ \Omega$

Current = $I = V / R = 10/31 = 0.32$ A

RESISTORS IN PARALLEL

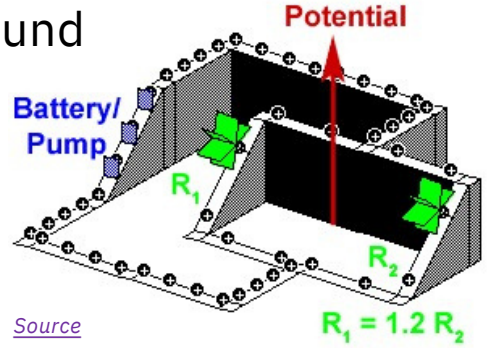
- When connected in parallel, the total conductance (GT) is found using

$$GT=G1+G2+G3+...$$

- and thetotalresistance is calculated using

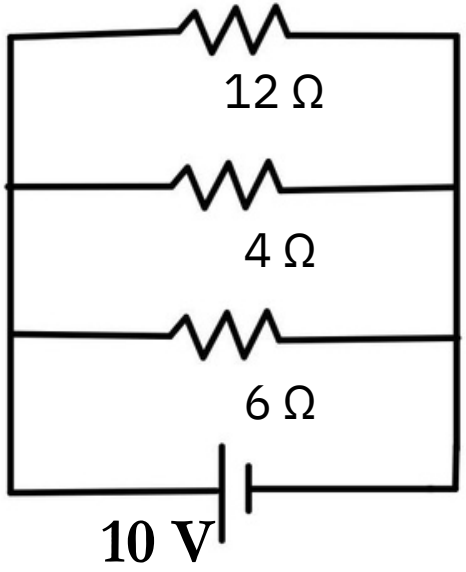
$$\frac{1}{RT} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R3} + ...$$

- Due to the reciprocal relationship, the total resistance is always smaller than any individual resistance
- Since there is more than one possible path, the current divides itself according to the resistance of each path



Source

RESISTORS IN PARALLEL



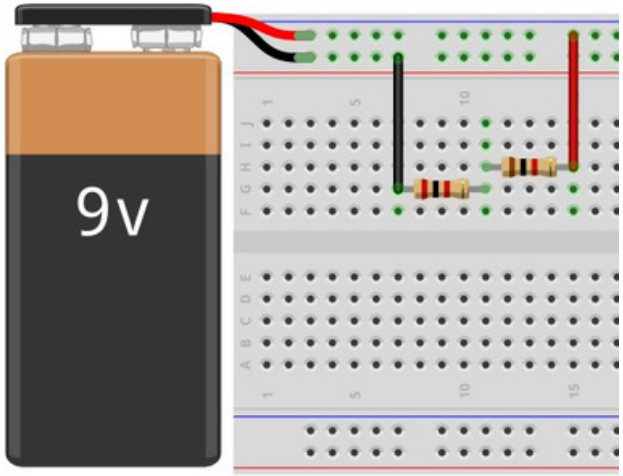
Example :

$$\frac{1}{R_T} = \frac{1}{12} + \frac{1}{4} + \frac{1}{6} \Rightarrow R_T = 2\ \Omega$$

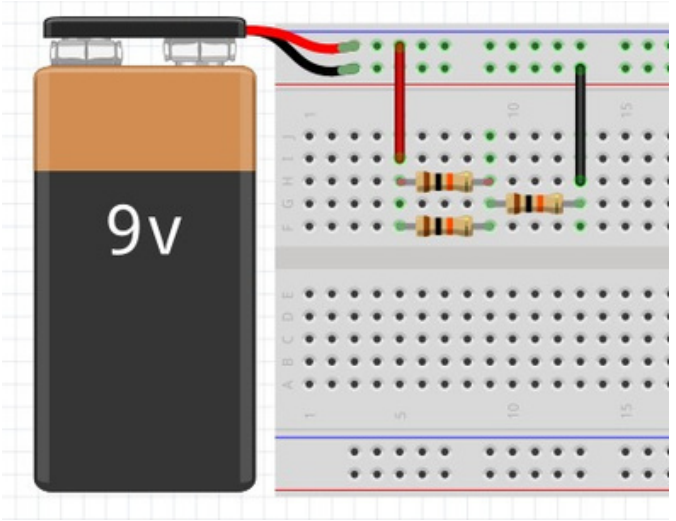
smallest resistor = more current passes
largest resistor = leastcurrent passes

ACTIVITY -IV

(I)



(II)



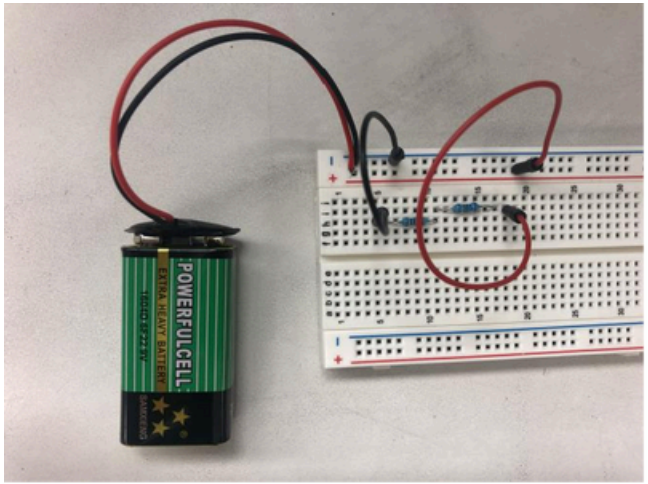
Determine the connection type and calculate total resistance for each figure

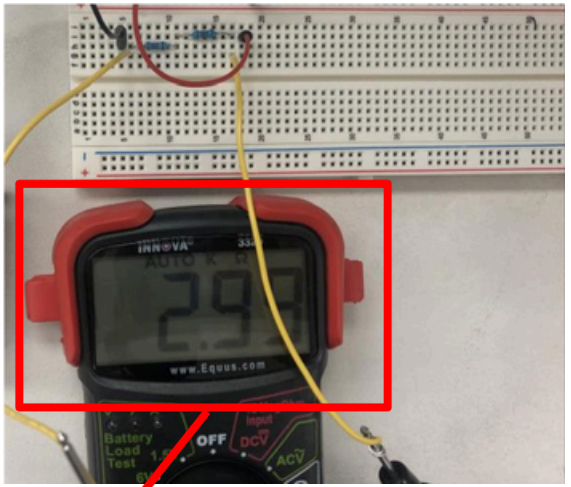
ACTIVITY –IV (SOLUTION)

(I)

Series Connection

R_T= 2000 + 1000 = 3 kΩ





2.99 kΩ

ACTIVITY –IV (SOLUTION)

(II)

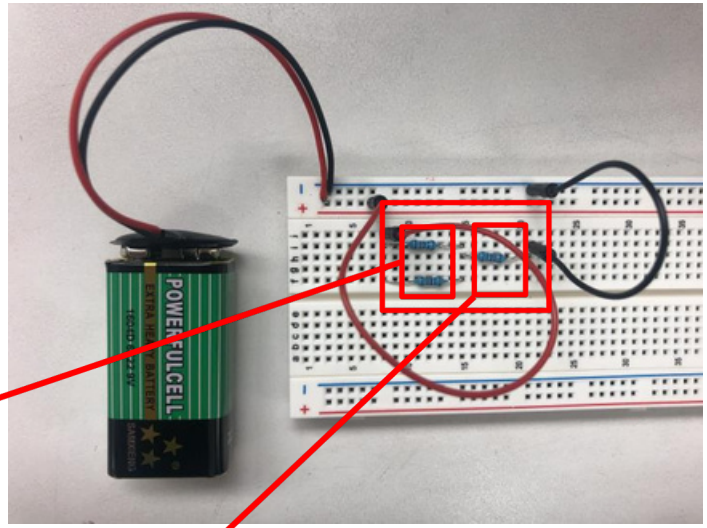
Parallel Connection

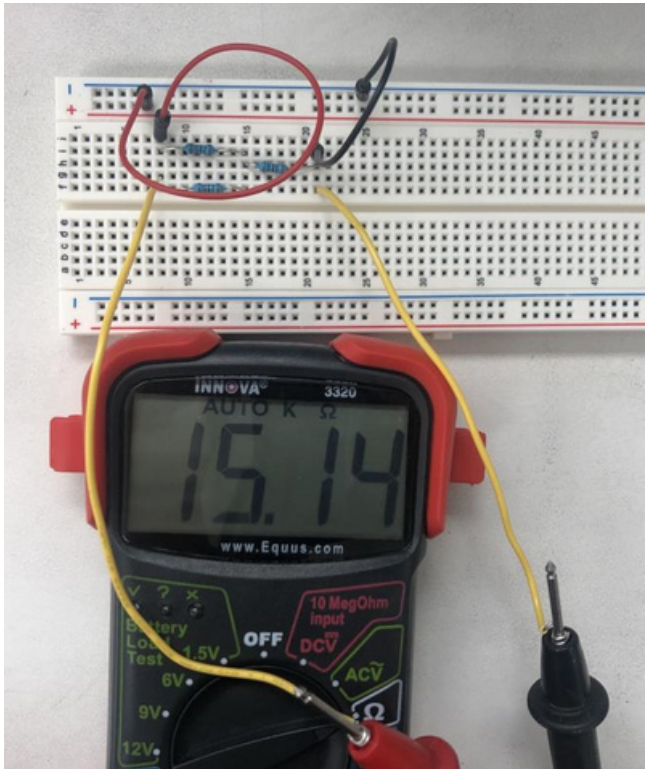
$$\frac{1}{R_P} = \frac{1}{10000} + \frac{1}{10000}$$

$R_P = 5000\ \Omega$

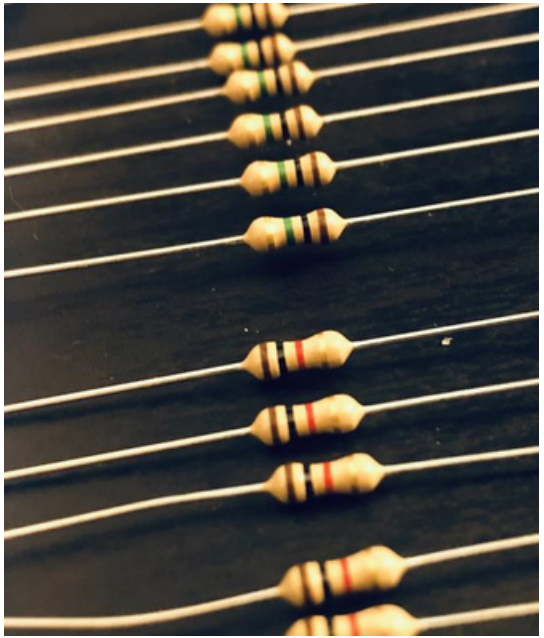
Series Connection

$R_T = 5000 + 10000 = 15\ k\Omega$

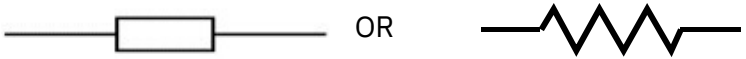


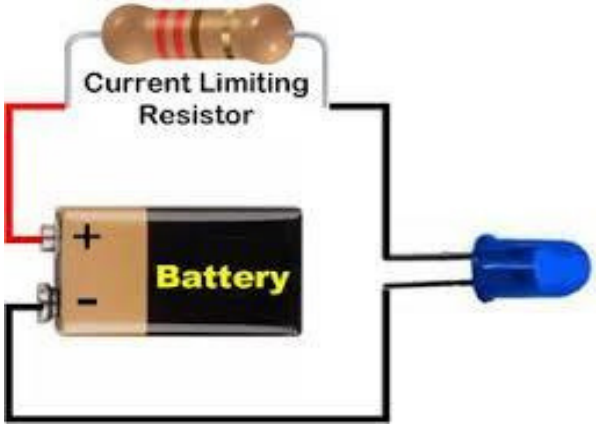


RESISTOR



Circuit symbol for a resistor (fixed resistance)





Source

A resistor is a passive component that limits the electrical current flowing in a circuit

CHOOSING THE RESISTOR

Supply Voltage = V1 = 9V LED

Voltage = V2 =3.6 V LED

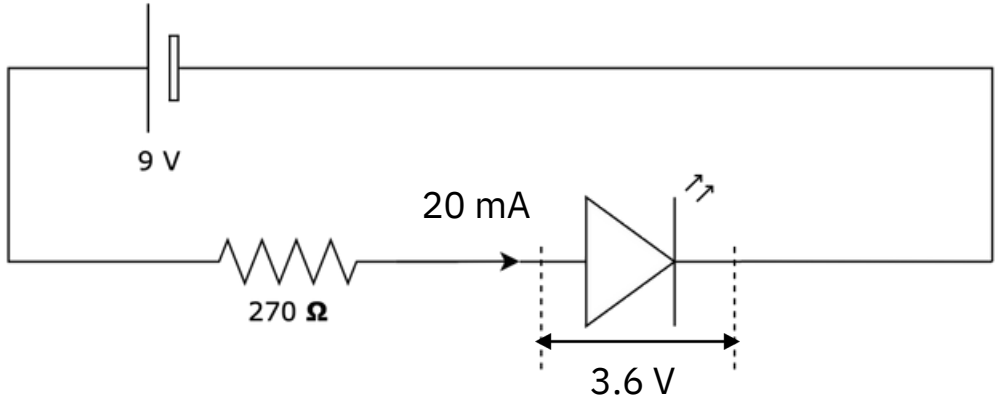
current = 20 mA = 0.02 A

V = I x R

→ $\vec{R} = \vec{(V2 - V1)} / I$

R = (9V –3.6V) / 0.02

R = 270 Ω

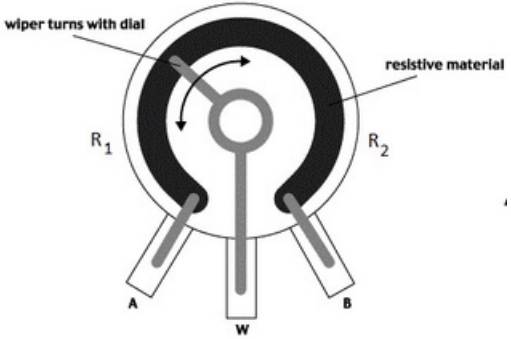


Any resistor with a value above 270Ωshould work

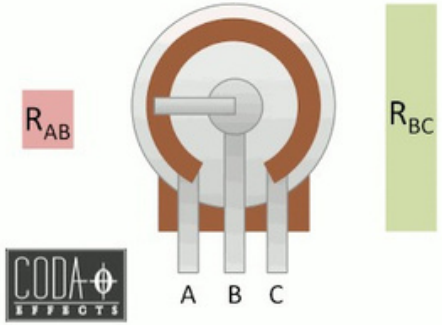
VARIABLE RESISTOR



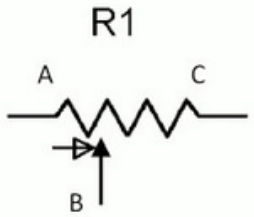
Source



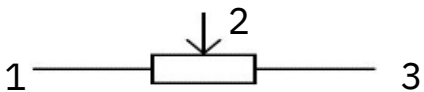
Source



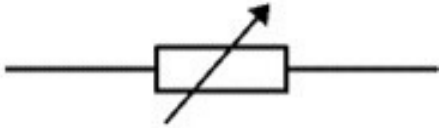
Source



Circuit symbol:



OR



OR



Source

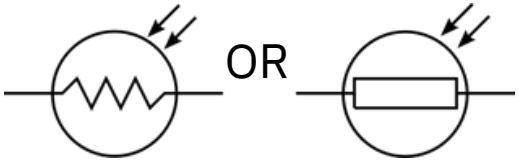
Variable resistors consist of a resistance track with connections at both ends and a wiper which moves along the track as you turn the spindle.

LIGHT DEPENDENT RESISTOR (LDR)



Source

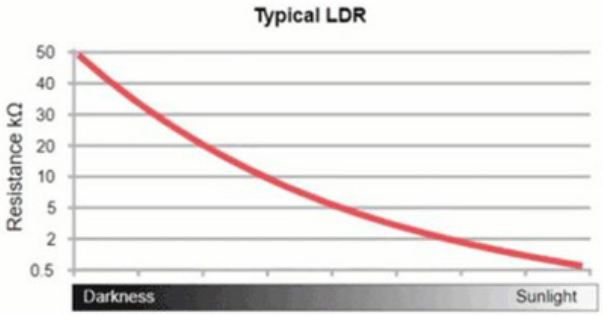
Circuit symbol:



Source

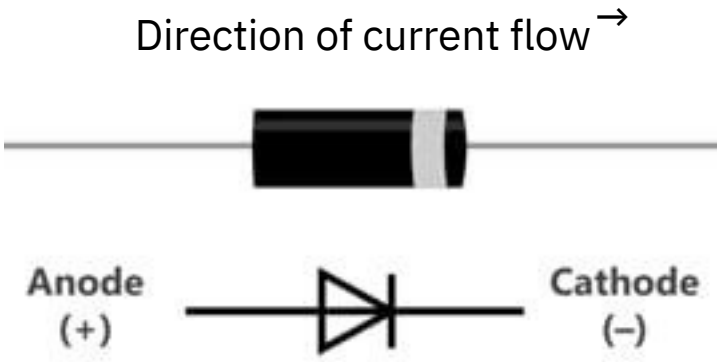


Source



- An LDR is an input transducer (sensor) which converts brightness (light) to resistance
- The resistance decreases as the brightness of light falling on the LDR increases

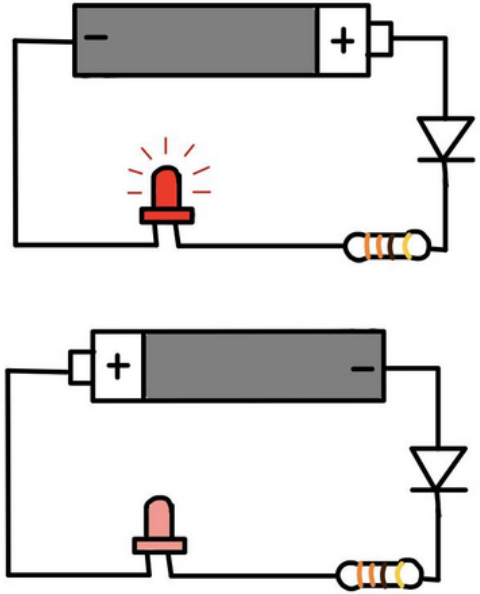
DIODE



Source

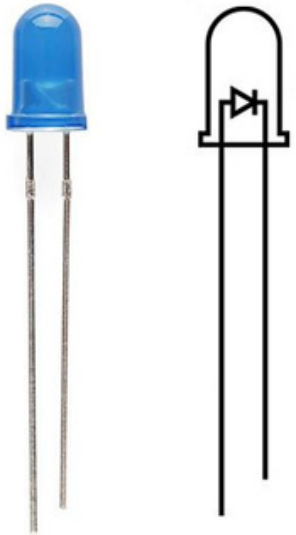


Source

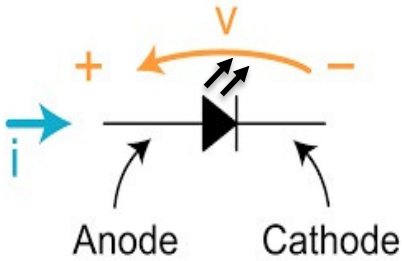


A diode is a two-terminal electronic component that conducts electric current in only one direction

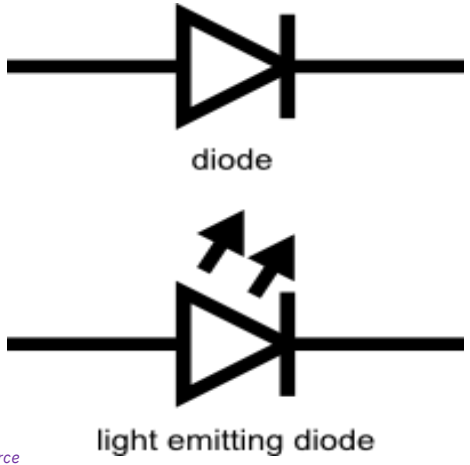
LIGHT-EMITTING DIODE (LED)



Source



Source

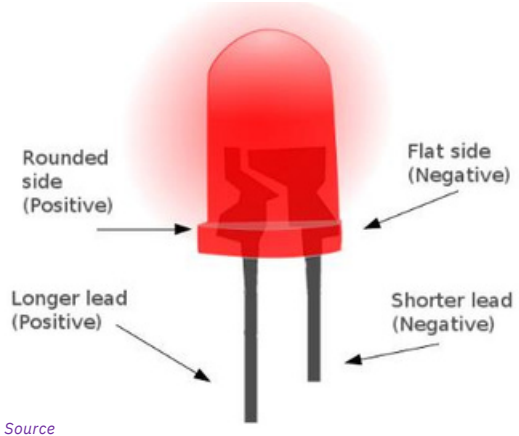


Source

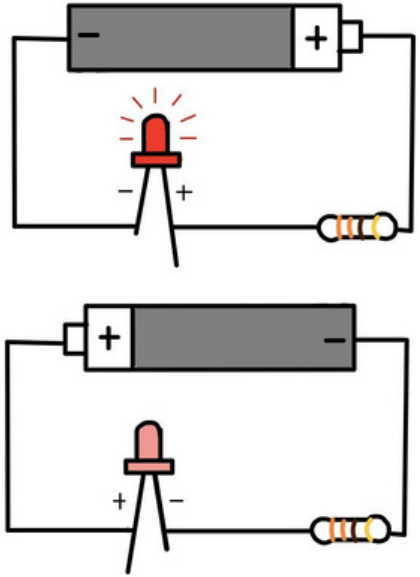
Circuit Symbol

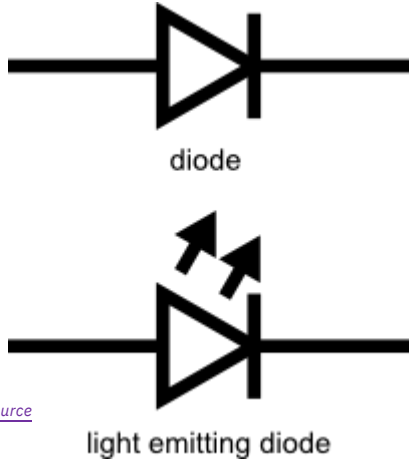
A Light emitting diode is simply a diode that lights up when current flows through one direction

LED POLARITY



Source



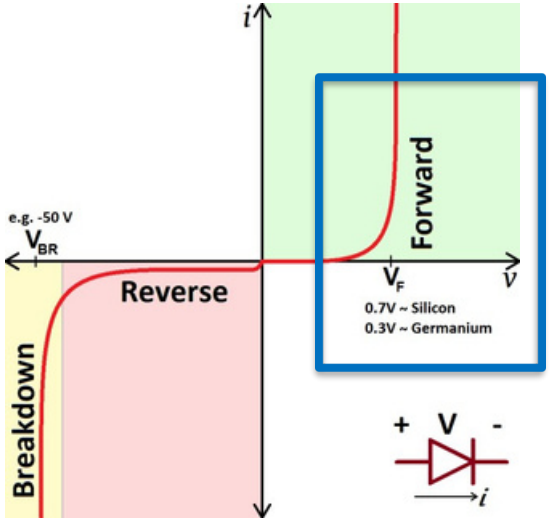


Source

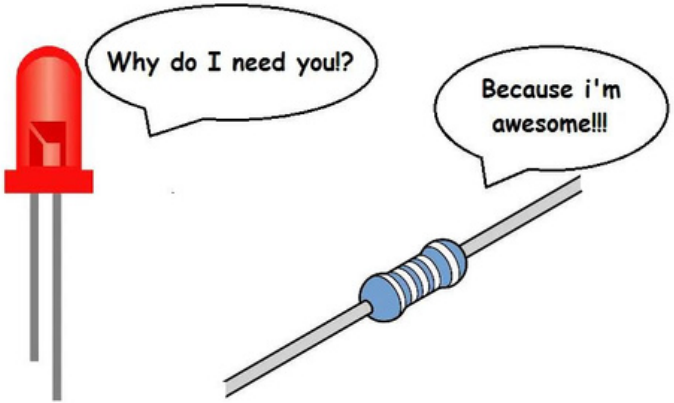
A Light emitting diode is simply a diode that lights up when current flows through one direction

DIODES AND LEDS

Never connect the LED or Diode circuit without resistor -Very dangerous –such a connection will burn the LED and other possible components in the circuit



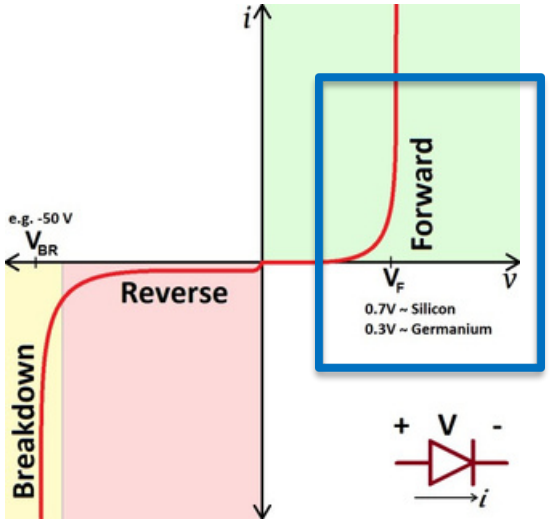
Source



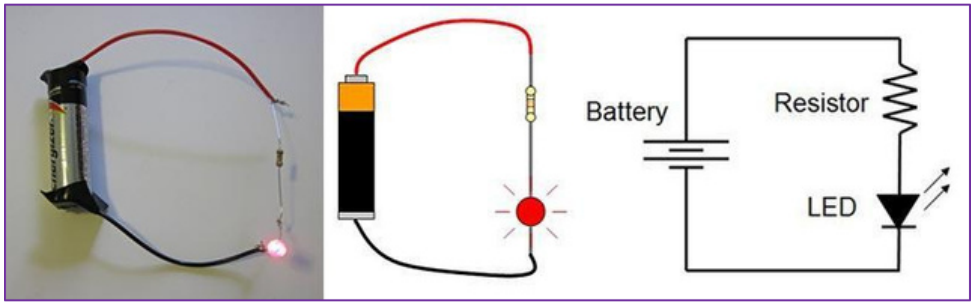
Source

DIODES AND LEDS

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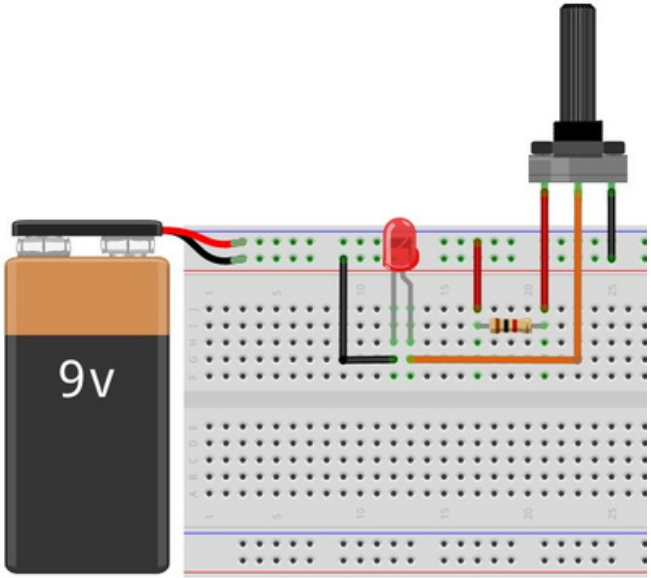
Source



Source

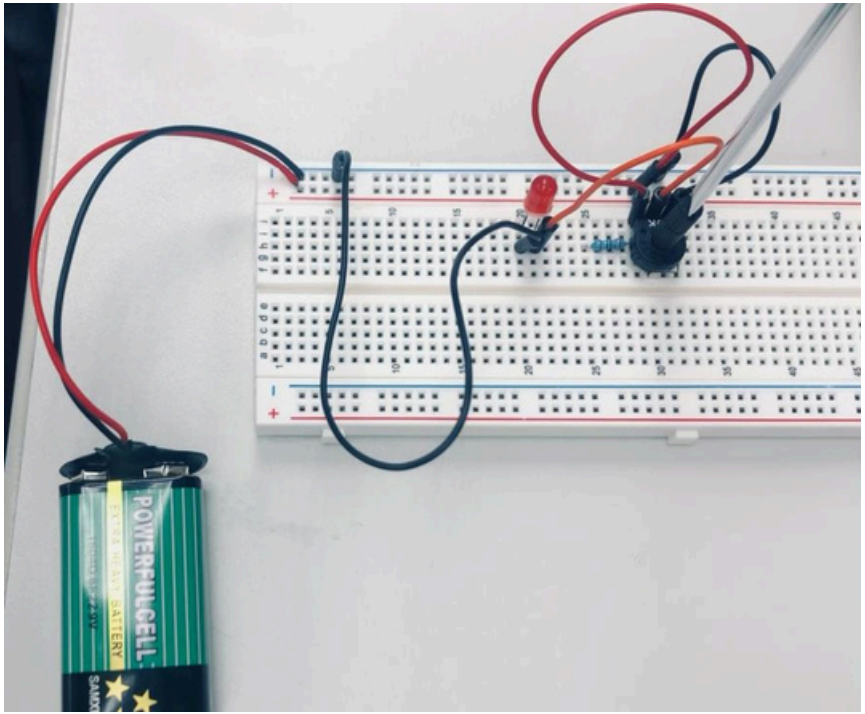
SAFE WAY TO USE LED

ACTIVITY -V



Construct this circuit and check if brightness of LED changes with variable resistor (10 kΩ)

ACTIVITY –V (SOLUTION)

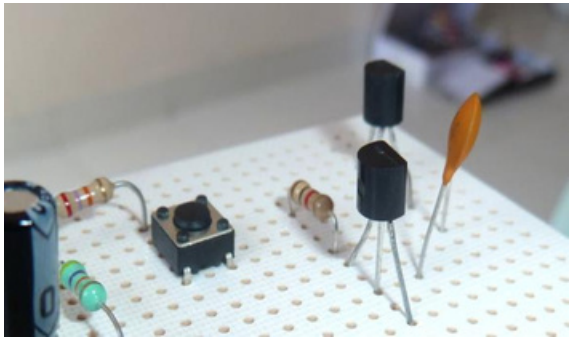


Video

Using variable resistor to control LED brightness

TRANSISTORS

- A transistor is like an electronic switch
- It can turn a current on and off
- A Bipolar Junction Transistor (BJT) has three pins: Base (b), collector (c), and emitter (e)
- BJT comes in two versions: NPN andPNP

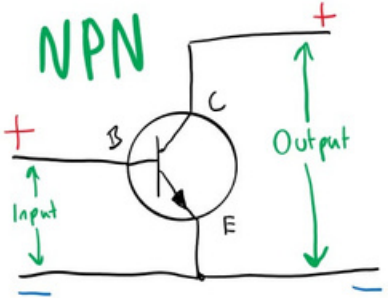


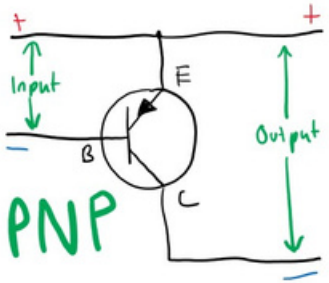
Source



1. Emitter
2. Base
3. Collector

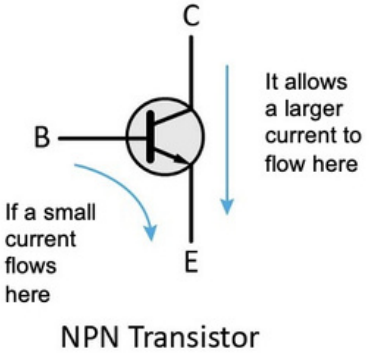
Source



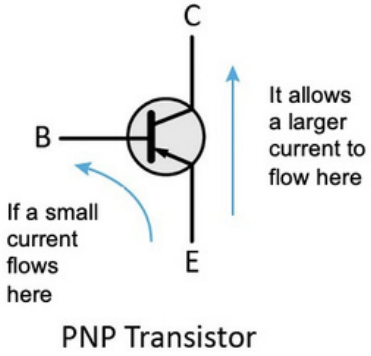


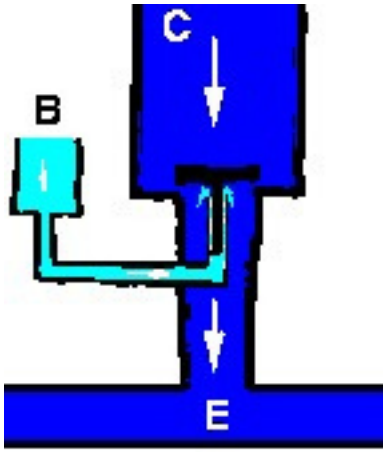
Source

- The transistor works because of the way p- and n-type semiconducting materials are sandw iched
- A current flowing from the base to the emitter “opens” the flow of current from the collector to the emitter



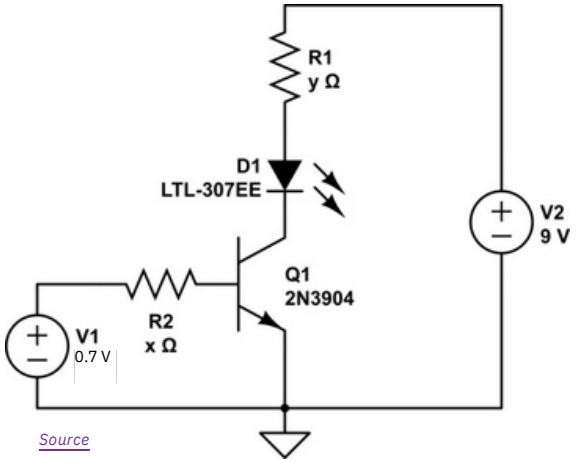
Source





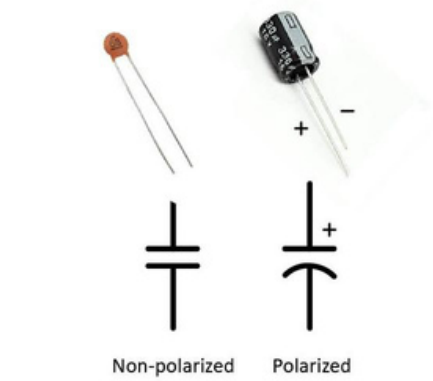
Source

Water analogy

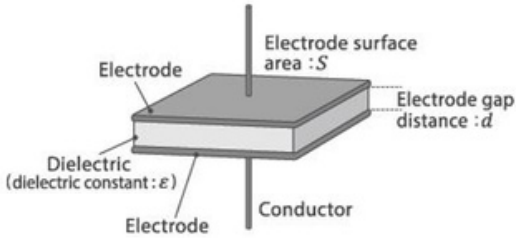


Source

CAPACITOR

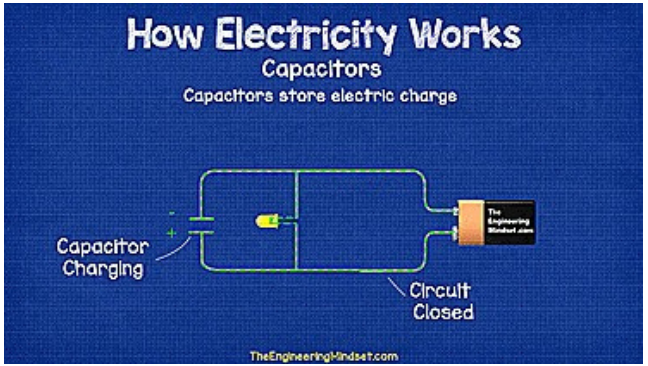


Source



Source

Capacitor in action



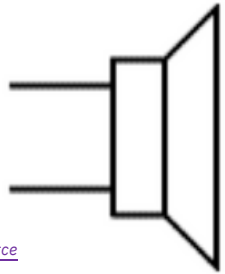
- A capacitor is a passive electronic component that stores energy in the form of an electrostatic field
- The capacitance is directly proportional to the surface areas of the plates and is inversely proportional to the separation between the plates
- Capacitance also depends on the dielectric constant of the substance separating the plates

SPEAKER

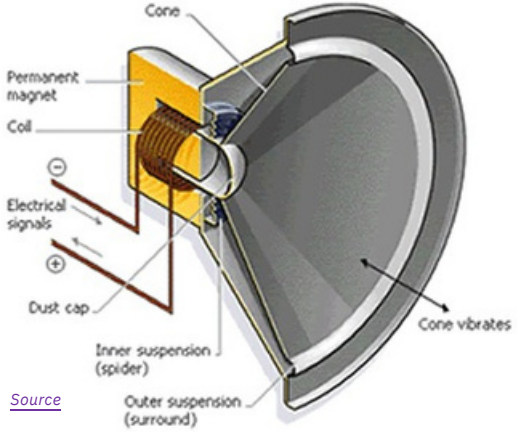


Source

Circuit symbol:

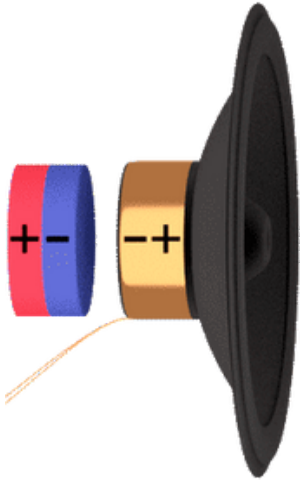


Source



Source

Speaker working



- A speakeris an electroacoustic transducer that converts an electrical signal into sound
- The speaker movesin accordance with the variations of an electrical signaland causes sound waves to propagate through a medium such as air or water

MECHANICAL SWITCHES

SWITCHES

SPDT



Source

DPDT



Source

DPST



Source

SPST

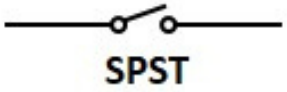


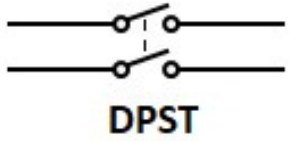
Source

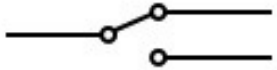


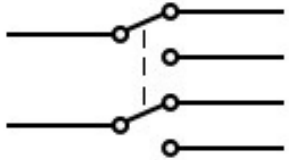
Source

Note: S –Single, D–Double, P –Pole, T -Throw





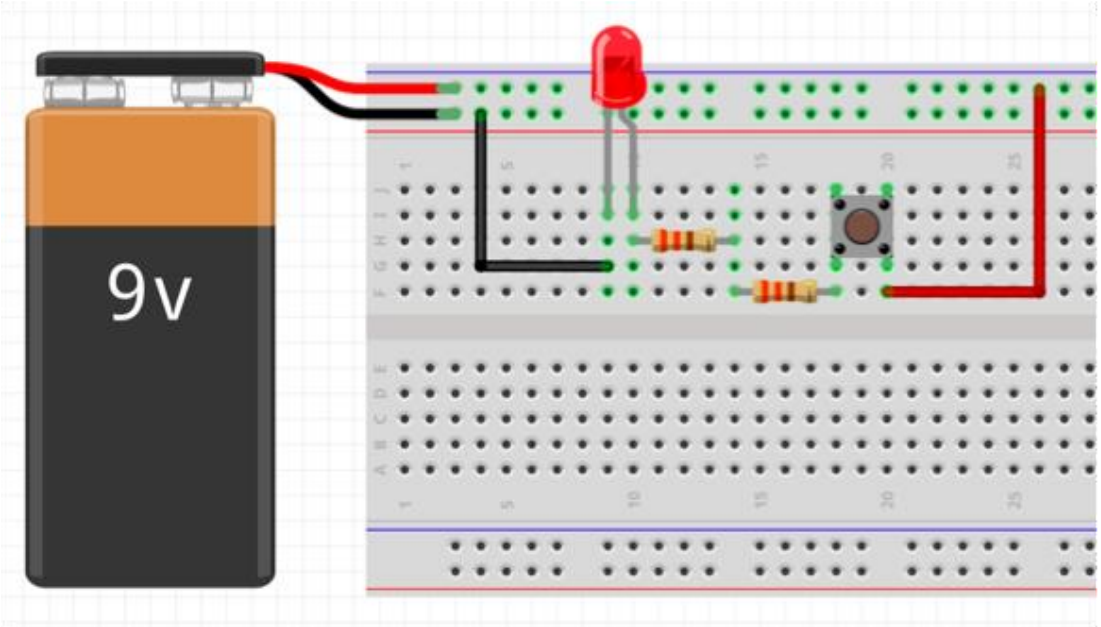




Source

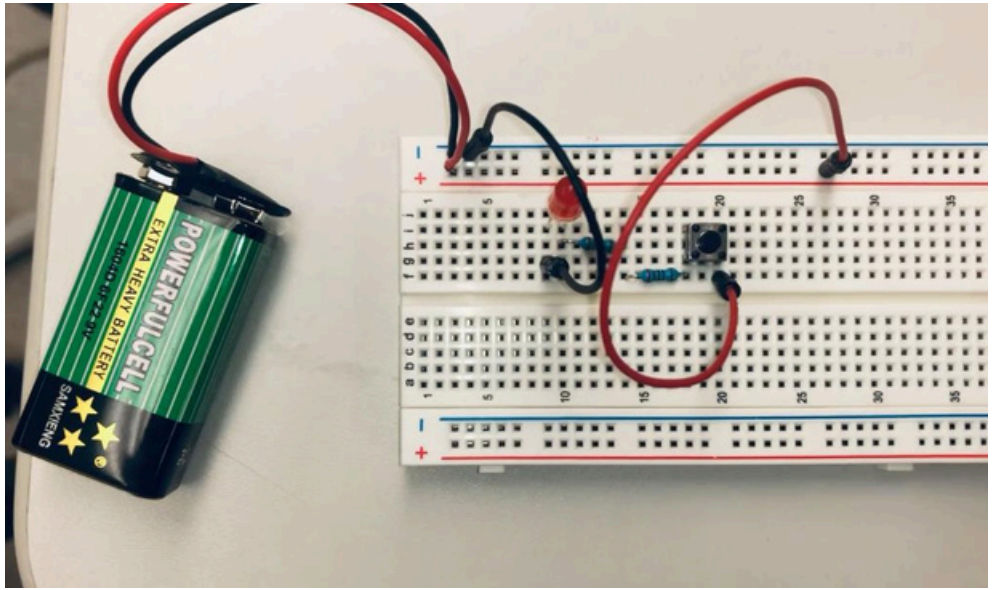
In electrical engineering, a switch is an electrical component that can "make" or "break" an electrical circuit, interrupting the currentor diverting it from one conductor to another

ACTIVITY -VI



Construct circuit such that switch can turn on and off LED

ACTIVITY -VI (SOLUTION)



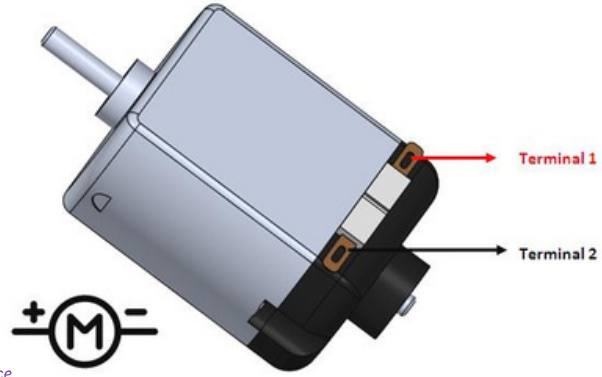
Video

Turning an LED on/off using a mechanical switch

MOTOR

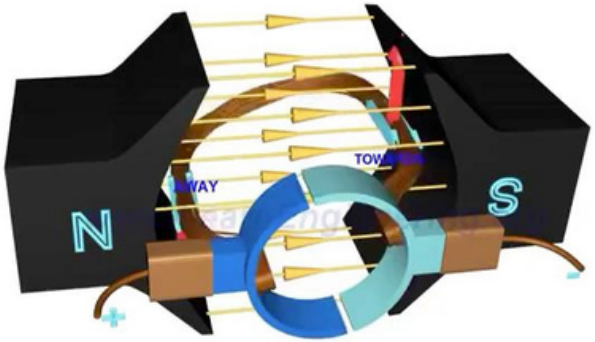


Source



Source

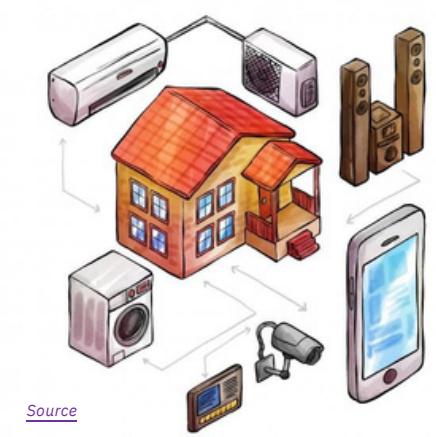
DC motor working:



- Motors are often used to turn wheels, gears, cams or pulleys
- Motors have two terminals that connect to a battery
- The directionof the turning depends on the way the battery wires are connected to the terminals

WHAT DIGITAL ELECTRONICS DO YOU USE?

- Computer
- CD & DVD players
- iPod
- Cell phone
- HDTV
- Digital cameras



Source



Source



Source



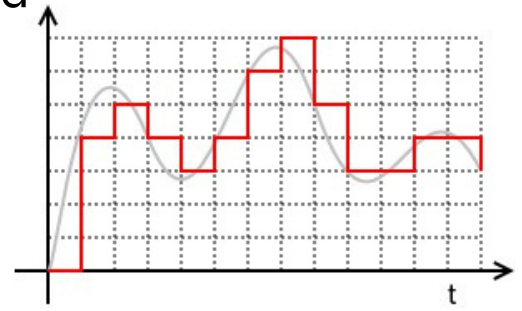
Source



Source

WHAT IS DIGITAL ELECTRONICS?

- Most signals in real life are not digital. For example, sound is an analogsignal
- On a CD, digital sound is encoded as 44.1 kHz, 16 bit audio
  - oThe original wave is 'sliced'44,100 times a second – and an average amplitude level is applied to each sample
  - o16 bit means that a total of 65,536 different values can be assigned, or quantizedto each sample
- DVD-Audio can be 96 or 192 kHz and up to 24 bits resolution
- To control things like switches we need true or false decisions, which is basically digital electronics



Source



Source

GEORGE BOOLE –Father of Boolean Algebra

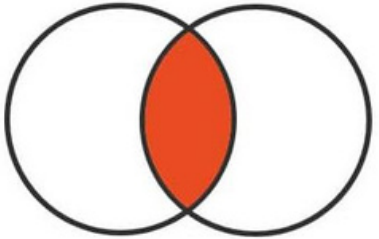
- George Boole, a British mathematician (1815-1864) proposed logic and math are equivalent

- All math functions can be determined using these 3 primary Boolean logic operators: AND, OR, and NOT

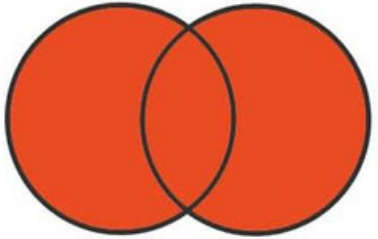
- AND narrows your search,

OR broadens your search, and

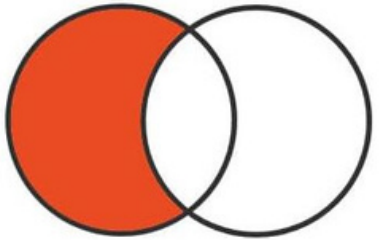
NOT is used to exclude concepts



AND



OR

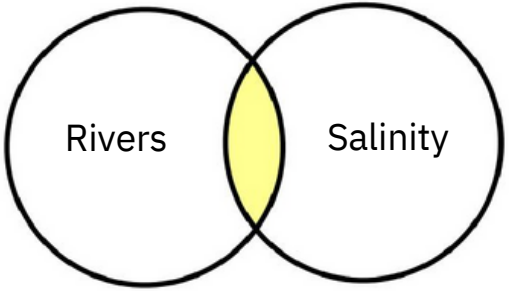


NOT

Source

THE AND OPERATOR (INTERSECTION)

River AND Salinity

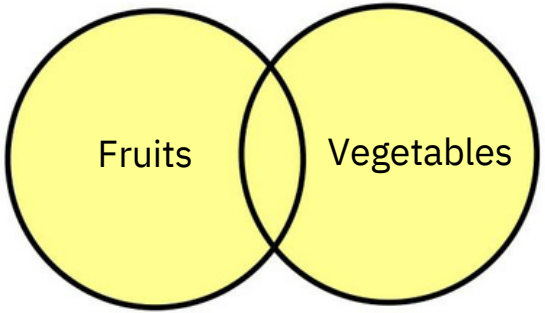


Dairy product AND Exported AND Europe

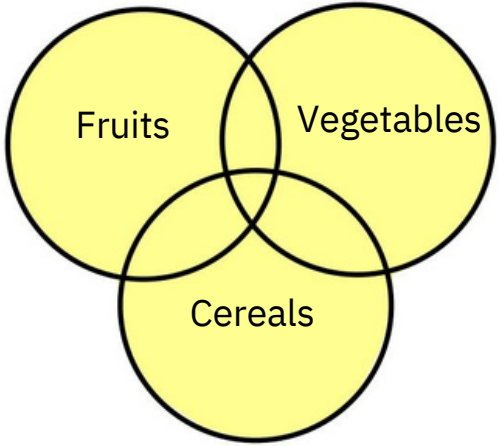


THE OR OPERATOR (EITHER, ANY)

Fruits OR Vegetables



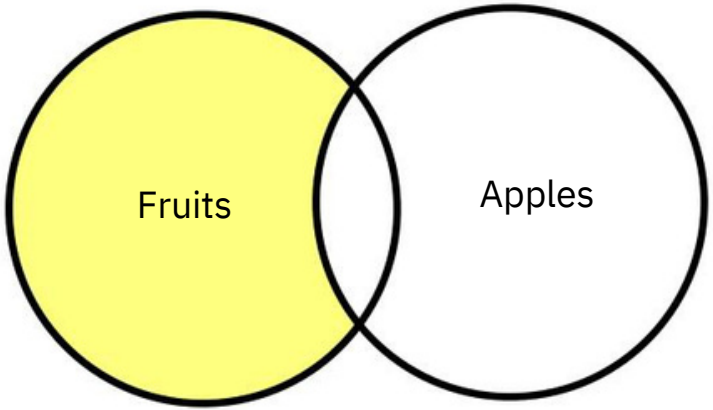
Fruits OR Vegetables OR Cereal



from CSIRO Australia

THE NOT OPERATOR

Fruits NOT Apples



Look at the Boolean Machine https://www.rockwellschrock.com/rbs3k/boolean/

BASIS FOR DIGITAL COMPUTERS

- The true-false nature of Boolean logic makes it compatible with binary logic used in digital computers

- Electronic circuits can produce Boolean logic operations

- Circuits are called gates
  - NOT
  - AND
  - OR

It's all binary at low-level



Source



Inverter
F = $\bar{A}$

A	F
0	1
1	0



AND
F = AB

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1



OR
F = A+B

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

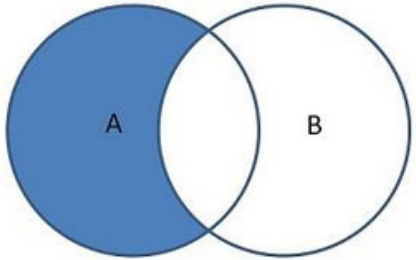
Source



Source

‘NOT’ GATE

- The simplest possible gate is called an "inverter ," or a NOT gate
- Onebit as input produces its oppositeas output
- The symbol for a NOT gate in circuit diagrams is shown below
- Thelogic table fortheNOT gate shows input and output



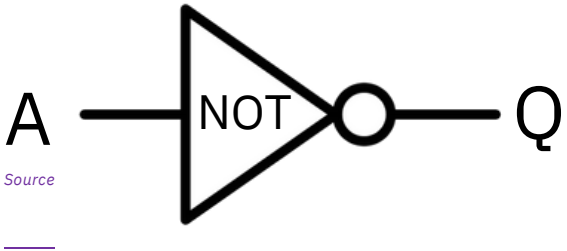
Source

A NOT B

A

0V

5V



Source

Q

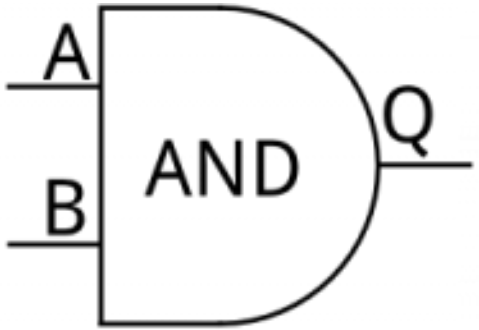
5V

0V

A	Q
0	1
1	0

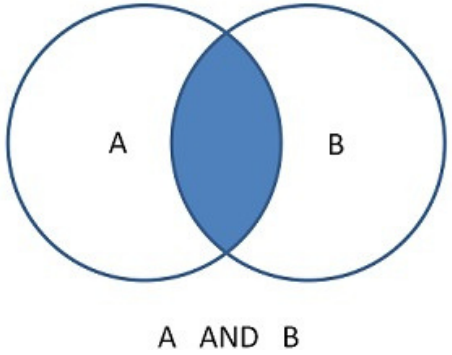
‘AND’ GATE

- The AND gate has the following symbol and logic table
- Two or more input bits produce one output bit
- Both inputs must be true (1) for the output to be true
- Otherwise, the output is false (0)



Source

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

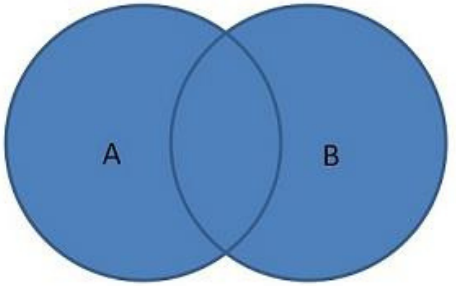


A AND B

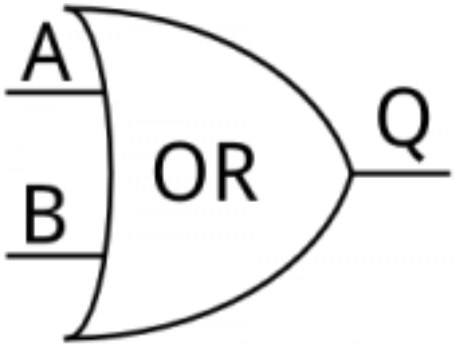
Source

‘OR’ GATE

- The OR gate has the following symbol and logic table
- Two or more input bits produce one output bit
- Either inputs must be true (1) for the output to be true



Source

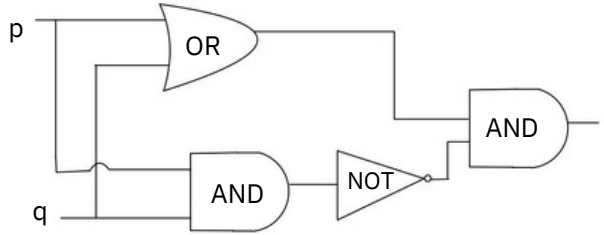


Source

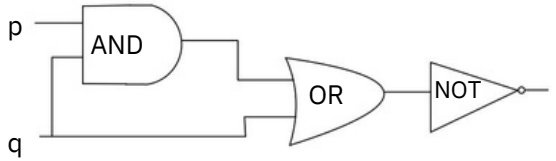
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

COMBINE GATES –EXERCISE 1

- Gates can be combined
- The output of one gate can become the input of another
- Try to determine the logic table for these circuits

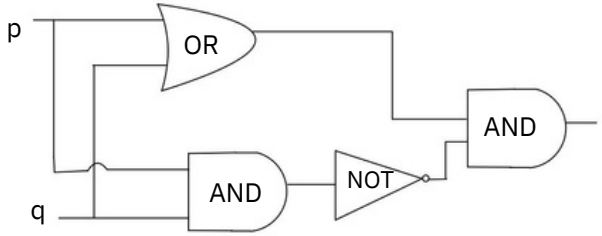


p	q	(p OR q) AND (NOT(p AND q))
1	1	
1	0	
0	1	
0	0	

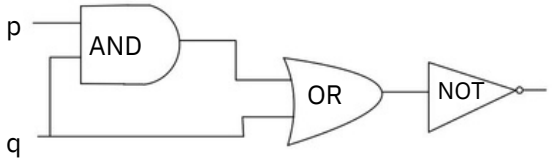


p	q	
1	1	
1	0	
0	1	
0	0	

EXERCISE 1-SOLUTION

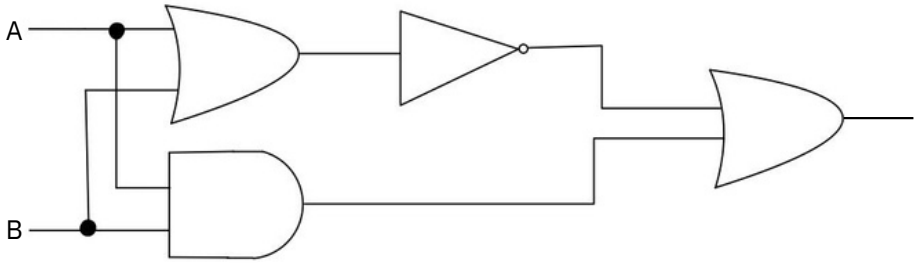


p	q	(p OR q) AND (NOT(p AND q))
1	1	0
1	0	1
0	1	1
0	0	0

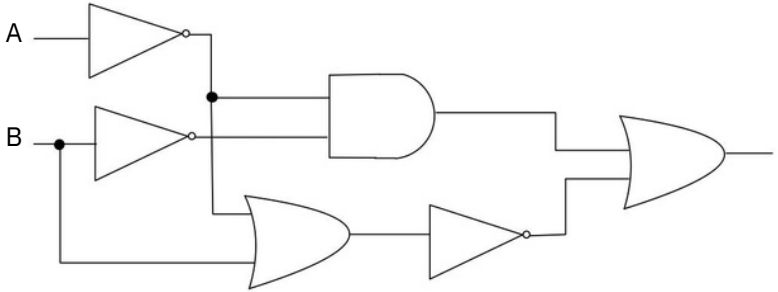


p	q	NOT ((p AND q) OR q)
1	1	0
1	0	1
0	1	0
0	0	1

EXERCISE -2

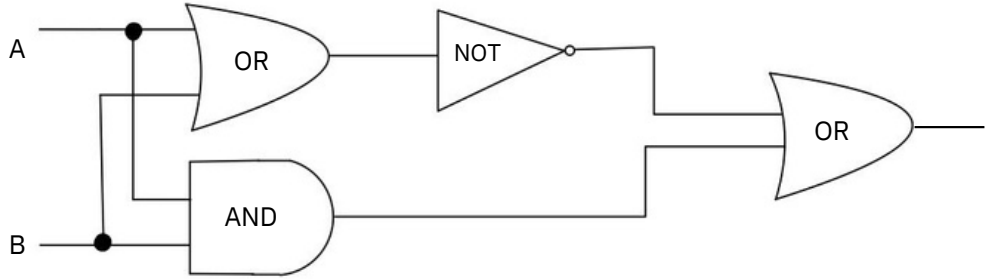


A	B	
1	1	
1	0	
0	1	
0	0	

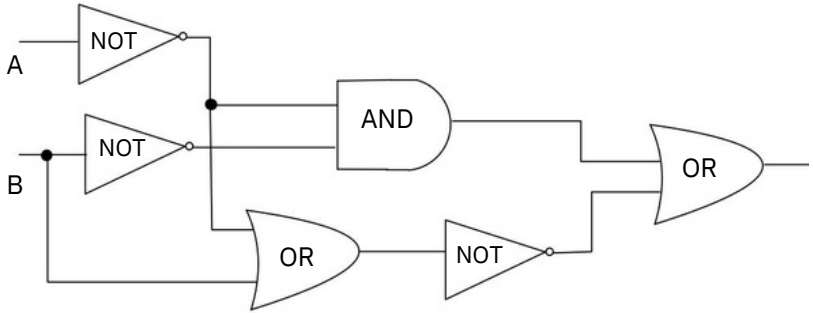


A	B	
1	1	
1	0	
0	1	
0	0	

EXERCISE -2 (SOLUTION)



A	B	NOT[A OR B] OR (A AND B)
1	1	1
1	0	0
0	1	0
0	0	1



A	B	(NOT[A] AND NOT[B]) OR (NOT{(NOT[A]) OR (B)})
1	1	0
1	0	1
0	1	0
0	0	1

CONCLUSION/CONTRIBUTIONS OF LESSON

- Most important concepts learned in: (i) lesson and (ii) task
- Contributions of lesson and task with respect to the entire PD program (only major/core contributions)
- How the information learned in this lesson will be beneficial for the next topic/lesson? (transition to next lesson)
- Important outcomes and findings?

Thank You!

Questions and Feedback?